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Bachelor Thesis

The Effect of Sentiment on Public Equities in Sweden

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Abstract

1 Introduction, Motivation and Purpose

In the efficient-market framework, persistent return predictability requires compensation for risk, market frictions, or information that prices have yet to incorporate (**??**). Similarly, the capital asset pricing model links expected returns to exposure to systematic risk (**?**). Within this benchmark, a systematic relation between investor sentiment and future cross-sectional returns requires either a risk-based interpretation or a pricing effect that arbitrage removes only slowly.

Behavioural finance provides a competing interpretation. If investors hold heterogeneous or biased beliefs, and if arbitrage is limited by short-sale constraints, financing risk, or model uncertainty, sentiment-driven demand can move prices away from fundamental values. Sentiment should have its strongest price effects among firms whose values are difficult to estimate and whose mispricing is costly to arbitrage. Baker and Wurgler formalize this prediction by arguing that sentiment should matter most for small, young, volatile, unprofitable, non-dividend-paying, intangible, distressed, or otherwise speculative firms (**??**). *The empirical problem is whether return differences across Swedish equities are consistent with rational risk compensation alone, or whether they also contain patterns consistent with sentiment-driven mispricing.*

Swedish evidence has mainly focused on constructing a local sentiment index or studying broad market-level relations. Zhang and Zhang construct a Swedish sentiment index using economic sentiment, consumer confidence, turnover, IPO activity, IPO returns, and an equity-to-debt proxy (**?**). Their work shows that the measurement logic in **?** can be adapted to Swedish data. A cross-sectional test remains open for Swedish firms with different exposure to valuation uncertainty and limits to arbitrage.

The empirical test constructs a Sweden-specific sentiment measure and links it to monthly return differences across characteristic-sorted portfolios. Beginning-of-period sentiment enters as a predictive state variable for subsequent relative returns. The cross-sectional design extends Swedish sentiment research from market-level index relations toward firm characteristics that theory identifies as sentiment-prone.

1.1 Research Question

> How does investor sentiment affect the cross-section of stock returns in the Swedish public equity market, and are the effects stronger for stocks with subjective valuations and greater limits to arbitrage?

1.1.1 Hypothesis

- **H1.** Beginning-of-period investor sentiment predicts relative future returns in the cross-section of Swedish equity returns.
- **H2.** The effect is stronger for stocks with subjective valuations or limits to arbitrage (**?**).

- **H0.** Beginning-of-period investor sentiment has a zero predictive relation with relative future returns.

2 Background

The OECD reports 909 listed companies in Sweden at the end of 2025, total market capitalisation of nearly USD 1.3 trillion, and market capitalisation equal to 194 percent of GDP. Among European peer countries, only the United Kingdom has more listed companies (**?**). That breadth makes Sweden empirically suitable for a cross-sectional sentiment test because the analysis compares firms across size, age, risk, profitability, dividend policy, tangibility, liquidity, turnover, idiosyncratic volatility, and sales growth.

The Swedish market also contains a clear split between established main-market firms and smaller growth-market firms. The OECD identifies Nasdaq Stockholm, Nordic Growth Market, and Spotlight Stock Market as the three main trading venues. Nasdaq Stockholm accounts for more than 99 percent of market capitalisation, while First North, NGM Nordic SME, and Spotlight contain many smaller and growth-oriented companies. Regulated markets are concentrated in industrial and financial companies, while MTFs have larger technology and health-care representation (**?**). The listed-firm distribution matches the prediction in **?** that sentiment effects should be more visible among firms with subjective valuations and stronger limits to arbitrage.

Public equity financing has also been unusually active in Sweden. From 2014 to 2025, the OECD reports 1,114 new listings and 615 delistings, giving a net increase of 499 listed companies. The median Swedish IPO size from 2000 to 2025 was USD 9 million, the smallest among the peer countries covered by the OECD (**?**). The large number of public firms outside the largest listed companies makes characteristic-sorted portfolio tests more informative than an aggregate index-only analysis.

SCB's shareholder statistics use a broader quoted-company classification. Statistics Sweden reports 935 quoted companies on Swedish marketplaces at the end of December 2025, including 128 firms on OMX Stockholm Large Cap, 132 on Mid Cap, 103 on Small Cap, 346 on First North, 128 on Spotlight, and 92 on NGM Nordic SME (**?**). The empirical universe used in this thesis is narrower because it applies common-equity, Swedish-ISIN, positive-price, and primary-security filters. The filtered Sweden panel contains 169,320 monthly security observations from January 2000 to December 2025 and 1,587 securities over the full sample. In December 2025, the filtered panel contains 900 selected primary common-equity lines.

2.1 OMXS30 Performance and Market Concentration

OMXS30 is Nasdaq Stockholm's leading blue-chip index and consists of 30 large and actively traded stocks. Nasdaq describes the index as suitable for derivatives and exchange-traded products because the underlying shares have high liquidity, and the index is revised twice a year (**?**). The index serves as a large-cap benchmark for the Swedish market, but it is narrower than this thesis sample because it excludes most small, young, and growth-market firms.

The monthly OMXS30 series used for the background analysis runs from January 2009 to December 2025. Over that window, the adjusted index level rises from 617.38 to 2,882.97, corresponding to a cumulative price return of 367.0 percent and an annualized price return of 9.5 percent. The same period contains several sharp market stress episodes. The largest peak-to-trough monthly drawdown is the 2021–2022 decline, where the index falls 24.4 percent from December 2021 to September 2022 and recovers its prior monthly high in February 2024. The next largest drawdowns are the 2015–2016 and 2011 episodes, both at 21.7 percent. The Covid-19 shock is shorter and sharper, with a 16.9 percent decline from January 2020 to March 2020 and recovery by September 2020.

Table 1: Largest OMXS30 Drawdowns in the Monthly OMXS30 Series

Peak month	Trough month	Recovery month	Peak level	Trough level	Drawdown (%)
Dec. 2021	Sep. 2022	Feb. 2024	2,419.73	1,828.98	-24.4
Feb. 2015	Jun. 2016	Oct. 2019	1,691.03	1,323.58	-21.7
Apr. 2011	Sep. 2011	Jan. 2013	1,162.84	910.17	-21.7
Jan. 2020	Mar. 2020	Sep. 2020	1,783.26	1,482.43	-16.9
Feb. 2025	Apr. 2025	Oct. 2025	2,724.70	2,434.06	-10.7

Note. Drawdowns are computed from monthly adjusted close values. Recovery month is the first later month in which the index reaches or exceeds the previous monthly peak. The monthly OMXS30 series covers January 2009 to December 2025.

The largest one-month drop in the same monthly series is March 2020, when OMXS30 falls 11.17 percent. Other large monthly declines occur in August 2011 at 10.44 percent, May 2019 at 9.88 percent, March 2025 at 8.46 percent, and June 2022 at 8.32 percent. The sample contains both slow valuation cycles and abrupt market-wide stress.

Market value in the filtered 2025 sample is concentrated among a small group of large companies. The ten largest companies account for 42.6 percent of sample market value, while the five largest account for 28.0 percent. Investor, Atlas Copco, and Volvo alone account for 20.8 percent after aggregating multiple listed share classes at company level. The concentration explains why an OMXS30-style large-cap index is not the same empirical object as this thesis cross-section. The index is dominated by liquid blue-chip firms, while this thesis tests whether sentiment predicts relative returns across the much broader Swedish equity sample.

Table 2: Largest Companies in the Filtered Sweden Sample, December 2025

Company	Sector	Market cap, SEK bn	Sample share (%)
Investor AB	Financials	1,011.8	8.74
Atlas Copco AB	Industrials	790.1	6.82
Volvo AB	Industrials	601.9	5.20
EQT AB	Financials	449.2	3.88
SEB AB	Financials	393.8	3.40
ASSA ABLOY AB	Industrials	378.7	3.27
Sandvik AB	Industrials	377.1	3.26
Swedbank AB	Financials	363.5	3.14
Saab AB	Industrials	287.0	2.48
Hexagon AB	Information Technology	284.2	2.45

Note. Market capitalisation is aggregated across listed share classes within company name. Sample share is measured relative to the filtered December 2025 Sweden sample.

Table 3: GICS Sector Composition, Sweden Sample and S&P 500

GICS sector	Sweden sample, Dec. 2025 (%)	S&P 500, Dec. 2025 (%)
Information technology	8.3	34.4
Financials	23.4	13.4
Industrials	39.0	8.2
Consumer discretionary	7.6	10.4
Health care	5.5	9.6
Communication services	3.9	10.6
Consumer staples	3.4	4.7
Real estate	4.9	1.8
Materials	3.9	1.8
Energy	0.0	2.8
Utilities	0.0	2.3

Note. Sweden values are market-capitalisation shares in the filtered Swedish equity sample using the latest available GICS sector classification. The Swedish sample also contains 0.1 percent unclassified market value, which is omitted from the table. S&P 500 values are index-level GICS sector weights reported by S&P Dow Jones Indices as of December 31, 2025 (**?**).

The U.S. benchmark is dominated by information technology and communication services, whereas the Swedish sample is more heavily exposed to industrials and financials. Sector composition affects the distribution of valuation uncertainty. A technology-heavy benchmark contains many firms whose value depends on growth options, intangible capital, and long-horizon cash-flow expectations. The Swedish sample combines a large number of small growth-oriented firms with aggregate market value concentrated in industrial and financial companies. The expected strength and location of sentiment effects may differ even if the underlying mechanism in **?** is the same.

At the end of 2025, SCB reports SEK 12,835 billion in market value for companies quoted on Swedish marketplaces. Households held 11.7 percent of that value, while foreign investors held 37.4 percent and financial corporations held 32.4 percent (?).[1] The U.S. market has a broader household-investor channel. Federal Reserve data report USD 47,186 billion in household directly held corporate equities, while the Investment Company Institute reports that 56 percent of U.S. households owned regulated funds and that nearly 130 million individuals owned regulated funds (??). JPMorganChase Institute evidence further shows that retail investment flows rose sharply from 2023 to early 2025, that early-2025 flows exceeded the pandemic-era peak in dollars transferred, and that younger and lower-income individuals now participate at much higher rates than in the mid-2010s (?).

The U.S. market has a more direct retail sentiment channel than Sweden. When individual investors are both numerous and active, sentiment can enter prices through marginal trading demand alongside long-run ownership shares. Sweden's smaller direct household ownership share points to a weaker retail-driven aggregate channel if professional and foreign investors correct mispricing more quickly. Retail participation can still affect prices through funds, attention-driven trading, IPO demand, and the marginal pricing of smaller and harder-to-arbitrage firms. ? show that retail trades are systematically correlated and that retail sentiment explains return comovement especially among small, low-priced, low-institutional-ownership, and costly-to-arbitrage stocks. ? show that individual investors are especially prone to buy attention-grabbing stocks. The Swedish setting points to sentiment effects that may be weaker at the aggregate market level and more visible in the cross-section among firms where retail attention and arbitrage frictions are more important.

3 Literature Overview

Miller's heterogeneous-expectations model shows that, when short selling is restricted, prices can be set by the optimistic investors willing to hold the available supply, with weaker influence from pessimistic investors outside the market (?). Blume and Easley add that, in incomplete markets, traders with inaccurate beliefs can survive through wealth dynamics, so competition can preserve heterogeneous beliefs (?). Uncertainty widens the dispersion of valuations. A young, volatile, newly listed, or hard-to-value firm can become overpriced even when the average forecast is moderate, because pessimistic investors can avoid the stock more easily than they can sell it short. The first empirical implication is that sentiment should be most visible where disagreement is large and negative views are costly to express.

Later limits-to-arbitrage work makes this mechanism dynamic. Noise-trader risk implies that rational arbitrageurs may avoid correcting mispricing when sentiment can push prices further away from fundamentals before convergence occurs (?). Shleifer and Vishny similarly argue that financing, agency, and performance-evaluation constraints allow recognized mispricing to survive (?). These papers sharpen Miller's asymmetry. Underpricing is easier to correct because investors can buy undervalued securities. Overpricing is harder to correct because it requires short selling, leverage, or a costly contrarian position. The cross-sectional prediction is that optimistic sentiment should produce relative overpricing in securities with high valuation uncertainty and high arbitrage risk.

[1] The Swedish 2025 ownership figures characterise the current market setting. The full 2000–2025 sample includes earlier ownership regimes and lower retail participation. Euroclear Sweden reports that 2025 was a record year for Swedish share ownership, with more than 2.8 million unique shareholders at year-end and an increase of approximately 47,300 shareholders during the year (?).

Baker and Wurgler define investor sentiment as a market-wide propensity to speculate and test whether this state predicts the cross-section of stock returns (??). The definition is deliberately operational. It requires a time-varying demand component that is weakly tied to fundamentals and that affects stocks differently depending on their valuation difficulty and arbitrage frictions. The empirical contrast is between sentiment-prone firms and more bond-like firms. Baker and Wurgler identify the sentiment-prone side as small, young, volatile, unprofitable, non-dividend-paying, intangible, distressed, or high-growth firms. The bond-like side consists of firms with more stable earnings, dividends, tangible assets, and longer histories.

In Baker and Wurgler's U.S. evidence, the predictive relation is conditional on the state of sentiment. After low sentiment, speculative firms tend to earn relatively high subsequent returns. After high sentiment, the same types of firms tend to earn relatively low subsequent returns (?). The return relation varies with sentiment because sentiment changes the relative valuation of firms whose cash flows are harder to anchor. The Swedish empirical object is the return spread between firms with different sentiment sensitivity.

Baker and Wurgler construct a composite sentiment index from several noisy proxies, including the closed-end fund discount, detrended turnover, IPO volume, average first-day IPO returns, the equity share in new issues, and the dividend premium (??). Each proxy has a separate economic interpretation. Turnover captures speculative trading activity, IPO variables capture hot-issue market conditions, the equity share captures issuance timing, and the dividend premium captures relative demand for safer dividend-paying firms. Baker and Wurgler's earlier evidence that the equity share in new issues predicts low future aggregate returns gives issuance behavior a separate market-timing foundation (?). The composite index extracts the common component across variables that are individually contaminated but jointly informative about speculative demand.

Baker and Wurgler first estimate a preliminary principal component from current and lagged proxy values, select the timing of each proxy by its correlation with the preliminary index, and then estimate the final index from the selected six variables (?). Their raw index loads negatively on the closed-end fund discount and dividend premium and positively on turnover, IPO volume, IPO first-day returns, and the equity share. The first principal component explains 49 percent of the variance in the raw proxies and 53 percent in the macro-orthogonalized proxies. Those figures show that the proxy set contains a sizeable common component, while the usual joint-hypothesis problem stays in view.

Qiu and Welch compare changes in consumer confidence and closed-end fund discounts with UBS/Gallup investor optimism (?). They find that changes in Michigan consumer confidence correlate strongly with changes in direct investor optimism, while changes in closed-end fund discounts are weakly related to the same optimism measure. For the Swedish setting, this supports the use of survey-based variables such as economic sentiment and consumer confidence. It also weakens a mechanical preference for U.S.-specific market proxies when those proxies are unavailable or less natural in a smaller European market.

Sentiment proxies can move with fundamentals as well as investor demand. Baker and Wurgler residualize their proxy variables against macroeconomic conditions before constructing an orthogonalized index (?). Finter, Niessen-Ruenzi and Ruenzi apply the same principle in Germany, where the country-specific sentiment indicator is built from macro-adjusted proxy variables before PCA

(?). Macro adjustment reduces business-cycle contamination and leaves residual variation that is more plausibly interpreted as sentiment. In Sweden, survey confidence, turnover, IPO activity, and capital-structure proxies can all respond to macroeconomic conditions.

Sentiment changes and sentiment levels answer different empirical questions. Baker and Wurgler use sentiment changes to study contemporaneous co-movement between sentiment and stock returns, while sentiment levels predict subsequent return reversals (?). Short-run sentiment movements are the relevant regressors for contemporaneous price pressure. Lagged sentiment levels are the relevant state variables for overpricing and later reversal. Ding, Mazouz and Wang formalize the same timing distinction in a multi-asset setting. Sentiment-prone assets react more strongly to short-run sentiment movements, while persistent high sentiment lowers their future expected returns because prices have already been pushed upward (?).

Stambaugh, Yu and Yuan sharpen the return-predictability test by focusing on overpricing. They study 11 anomaly strategies and show that anomaly profits are stronger after high sentiment, with the effect driven primarily by the short legs (?). The pattern follows Miller's short-sale asymmetry. If optimistic sentiment creates overpricing, the subsequent return effect should be strongest among securities identified as likely overpriced. Their later work on idiosyncratic volatility extends the mechanism. High idiosyncratic volatility increases arbitrage risk, and the negative idiosyncratic-volatility return relation is strongest among overpriced stocks (?). For this thesis, idiosyncratic volatility proxies for the difficulty of correcting sentiment-driven mispricing.

Persistent sentiment predictors can generate misleading return-predictability results if many portfolios and horizons are searched. Stambaugh, Yu and Yuan address this concern directly by comparing investor sentiment with random persistent predictors and show that the coherent anomaly pattern is difficult to reproduce by chance (?). Swedish evidence is more persuasive when signs line up across theoretically related portfolios. It is stronger when effects concentrate among difficult-to-arbitrage firms and when long-leg and short-leg results match the overpricing mechanism.

Baker, Wurgler and Yuan show that sentiment indices can be constructed outside the United States and that country sentiment can be decomposed into global and local components (?). Local sentiment is especially relevant for within-country cross-sectional returns, while global sentiment is more important for broad market returns. A Swedish sentiment index can be treated as a local state variable for Swedish cross-sectional tests, while global and Nordic spillover channels remain separate extensions unless they are explicitly modeled.

Finter, Niessen-Ruenzi and Ruenzi construct a German sentiment indicator from survey, options, trading, fund-flow, IPO, and issuance-related proxies for German stocks from 1993 to 2006 (?). Their macro-adjusted PCA index explains contemporaneous return spreads between sentiment-sensitive and sentiment-insensitive stocks, especially for high-volatility, high-idiosyncratic-risk, young, small, non-dividend-paying, and unprofitable firms. However, they find only weak evidence that the index predicts future return spreads. The German evidence prevents the Swedish thesis from importing the U.S. reversal evidence as a foregone conclusion. In a European market, sentiment may be measurable and economically meaningful even if predictive reversals are weaker, shorter-lived, or more dependent on specification.

Shoaib studies Nordic sentiment and U.S. spillovers, which provides regional context for local and foreign sentiment channels (?). Zhang and Zhang construct a Swedish investor sentiment index from economic sentiment, consumer confidence, turnover, IPO volume, IPO returns, and an equity-to-debt proxy over January 2014 to December 2022 (?). Their contribution is methodological. They show how the proxy family in ? can be adapted to Swedish data availability. Their empirical test is mainly a market-level relation with Swedish broad indices, leaving relative returns across sentiment-sensitive firms for further analysis.

Prior work explains why sentiment should affect prices, how a latent sentiment index can be measured, why the effect should be strongest among hard-to-value and hard-to-arbitrage firms, and why local sentiment is relevant outside the United States. What remains less explored is whether these mechanisms appear in the cross-section of Swedish equity returns. The present analysis addresses that gap by constructing a Sweden-specific sentiment measure and testing whether lagged sentiment predicts return differences between Swedish firms with different exposure to valuation uncertainty and limits to arbitrage.

4 Method

The empirical design proceeds from raw Swedish market, accounting, sentiment, macroeconomic, and IPO data to monthly sentiment-conditioned return tests. Investor sentiment is latent, accounting variables are observed with delay, and return predictability must be evaluated using information available before the return is measured. The method separates sample construction, source resolution, point-in-time accounting, characteristic construction, sentiment-index construction, portfolio formation, and regression testing.

4.1 Empirical Scope and Unit of Observation

Market and accounting data cover Swedish listed firms where Compustat market and fundamental records are available, with the broad data backbone running from January 2000 to December 2025 and the Swedish sentiment-paper sample concentrated on January 2014 to December 2022. The main return and portfolio unit is the security-month, while accounting variables are linked at the company level. The issuer identifier, company_id, is used for issuer-level fundamentals, while the share-class identifier, security_id, is used for prices, returns, and portfolio assignment.

company_id is defined as country code plus gvkey, and security_id is defined as country code plus gvkey plus iid. These identifiers determine how market data, accounting data, characteristics, and portfolios are joined. Because the sentiment literature tests cross-sectional predictions across assets, this thesis interprets return patterns at the security-month level while using company-level accounting information where fundamentals are issuer-specific.

4.2 Data Sources

The analysis combines Swedish market, accounting, sentiment, IPO, interest-rate, and macroeconomic data into monthly variables. Daily market data provide trading-day prices, shares, turnover, total-return factors, and IPO first-close matching. A monthly market file supplies the preferred post-2007 monthly price source. Quarterly fundamentals provide accounting variables and filing dates. Survey sentiment is drawn from the economic and consumer sentiment workbook. IPO offer prices and dates are drawn from the Bloomberg IPO file. Interest-rate controls are drawn from Swedish 3-month interbank and 10-year yield files. Macro controls are drawn from CPI, PPI, industrial production, and employment series. The constructed variables are grouped into sentiment proxies, firm characteristics, and controls.

4.3 Sample Construction and Security Universe

The security universe is formed from Swedish daily prices, monthly prices, and quarterly fundamentals, with company and security identifiers assigned before returns and characteristics are constructed. Return, factor, and IVOL construction use a common-equity screen based on available issue-type fields. The screen retains ordinary equity observations where the source identifies the instrument as common equity or where available issue-type information supports inclusion. The usable sample varies by empirical layer because market data, fundamentals, sentiment proxies, macro controls, and IPO observations have different coverage. Each empirical test reports the broad source universe and the narrower sample used for that test.

4.4 Market Data Construction

Monthly market observations are constructed through a hybrid source-resolution rule. Before 2007, month-end observations are derived from the last available daily trading record in each month. From January 2007 onward, the monthly workbook is the preferred source when it provides a valid price, while daily-derived observations are retained as fallback when the monthly source is missing or invalid. Monthly returns are computed from adjusted month-end prices using the total-return factor, and market equity is measured as month-end price multiplied by shares outstanding. The resolved monthly market panel is the return backbone for characteristics, factors, portfolio sorts, and later regressions.

4.5 Accounting Data and Point-in-Time Fundamentals

Accounting information is converted from quarterly fundamental records into filing-level observations and then into monthly point-in-time snapshots. A report is treated as available on the final filing date where available, otherwise on the preliminary filing date, and otherwise after a fallback lag from fiscal period end. Quarterly observations use a 90-day fallback lag and semiannual observations use a 180-day fallback lag. For each month-end, the snapshot table assigns the latest available filing whose effective month-end is observable by that date. This means accounting variables enter portfolio and characteristic construction only after they could have been known.

Point-in-time accounting snapshots prevent book equity, profitability, dividends, investment, and growth variables from using information unavailable at portfolio formation. The characteristics used to identify hard-to-value and difficult-to-arbitrage firms follow **?** and are measured using information observable at the monthly portfolio date.

The raw Compustat extracts also impose important limits on which accounting characteristics from **?** can be measured in Sweden. **??** reports raw field coverage before any monthly carry-forward is applied. Price, return-factor, and shares-outstanding fields are strongly covered in the market files, while daily trading volume is less complete. Balance-sheet fields needed for total assets, book-equity repair, and liabilities are broadly available, while revenue and tangible fixed-asset fields are usable but incomplete. revtq is used as the single revenue input for the sales-growth variable, GS, rather than mixing alternative sales fields. Earnings coverage is strong when income before extraordinary items is used, but the alternative net-income fields are essentially unavailable in the extract. Dividend information is present only for a minority of raw firm-quarter records, which is why dividend-payer status is based on observed dividend history rather than a single-period dividend ratio. External-finance variables are substantially weaker, with sale of common and preferred stock present in fewer than one third of firm-quarters, while repurchases and long-term debt issuance have very low coverage. Research and development expense and deferred-tax adjustment fields are not present in the saved Compustat extract. Consequently, this thesis excludes RD/A and does not use EF/A as a main characteristic, while retaining the better-supported accounting variables for book-to-market, earnings/profitability, dividend-payer status, asset tangibility, and revenue growth.

4.6 Firm Characteristics

? and **?** predict that sentiment effects concentrate in firms whose valuations are subjective and whose mispricing is costly to arbitrage. The Swedish characteristics translate that prediction into monthly sorting variables. **?** sharpen the same logic by arguing that sentiment-related mispricing should appear most clearly on the overpriced side of anomaly strategies, because pessimistic investors face greater constraints when correcting overpricing than optimistic investors face when correcting underpricing. The Swedish characteristic set separates firms by valuation uncertainty, cash-flow anchoring, trading frictions, and arbitrage risk.

4.6.1 Market Equity

$ME_{i,t}$ is measured as month-end price multiplied by shares outstanding and scaled to market equity. The small-firm portfolio is the sentiment-prone side of the size sort. Small firms are usually followed by fewer investors, have less diversified operations, and can be more expensive to arbitrage. In the Swedish portfolio tests, the directional size spread is coded as small firms minus large firms.

4.6.2 Age

$age_{i,t}$ is measured as the number of months since the first observed listing month divided by twelve. Young firms are the sentiment-prone side because their operating histories are shorter and their valuations rely more heavily on expectations about future growth. The directional age spread is coded as young firms minus old firms.

4.6.3 Total Risk

$risk_{i,t}$ is the rolling standard deviation of monthly percentage returns over the previous twelve months, with at least nine monthly observations required. High total risk is used as a proxy for valuation uncertainty and disagreement. The risk spread is coded as high-risk firms minus low-risk firms.

4.6.4 Book-to-Market

$BE/ME_{i,t}$ is measured as repaired point-in-time book equity divided by market equity. Book equity must be positive and market equity must be positive. The variable is retained as a valuation diagnostic because book-to-market is closely related to the dividend-premium and valuation channels in the sentiment index. The main reported tests inspired by **?** emphasize the validated sentiment-prone characteristics rather than treating $BE/ME_{i,t}$ as an independent source of evidence.

4.6.5 Profitability and Unprofitable Firms

Earnings are measured from the trailing point-in-time accounting snapshot. $E^+/BE_{i,t}$ scales positive trailing earnings by book equity, with positive book equity required. $UNPROFITABLE_{i,t}$ equals one when trailing earnings are non-positive and equals zero when earnings are positive. Low profitability and non-positive earnings identify firms with weaker cash-flow anchors. The unprofitable-firm spread is coded as unprofitable firms minus profitable firms, while the continuous profitability spread is coded as low profitability minus high profitability.

4.6.6 Dividend-Payer Status

$D\_PAYER_{i,t}$ records regular dividend-payer status from the point-in-time fundamental layer, with daily dividend evidence used as a fallback where available. $NON\_D\_PAYER_{i,t}$ equals one minus $D\_PAYER_{i,t}$ when dividend status is observed. Non-dividend payers are the sentiment-prone side because their valuation depends more on distant cash flows and less on current payout evidence. The dividend-status spread is coded as non-dividend payers minus dividend payers. Dividend-status evidence receives a mechanical-overlap caveat because $SENT_t^{\perp}$ includes $DIVP_t$.

4.6.7 Asset Tangibility

$PPE/A_{i,t}$ is measured as gross property, plant, and equipment divided by total assets and scaled by 100, with positive total assets required. Low tangibility is the sentiment-prone side because firms with fewer observable hard assets have weaker balance-sheet anchors for valuation. The tangibility spread is coded as low tangibility minus high tangibility.

4.6.8 Sales Growth

$GS_{i,t}$ follows the growth-opportunity characteristic in **?**. It is measured as the year-over-year change in trailing-twelve-month revenue divided by prior-year trailing-twelve-month revenue,

$$GS_{i,t} = \frac{REVTTM_{i,t} - REVTTM_{i,t-12}}{REVTTM_{i,t-12}}. \tag{1}$$

The Swedish sales-growth construction uses revtq as the single production revenue input. $GS_{i,t}$ is computed only when prior-year trailing revenue is positive and current trailing revenue is non-negative. High sales growth is the sentiment-prone side because high-growth firms rely more heavily on distant cash-flow expectations and growth opportunities. The sales-growth spread is coded as high sales growth minus low sales growth.

4.6.9 Illiquidity

$ILLIQ_{i,t}$ is an Amihud-style monthly illiquidity measure. It is computed as the monthly average of absolute cleaned daily returns divided by daily traded value in millions, with at least five usable daily observations required. High illiquidity is the sentiment-prone side because trading frictions make mispricing harder to correct. The illiquidity spread is coded as high illiquidity minus low illiquidity.

4.6.10 Share Turnover

$XTURN_{i,t}$ is measured as monthly traded shares divided by month-end shares outstanding. Low turnover is coded as the sentiment-prone side in the portfolio tests. The interpretation is empirically ambiguous because turnover can proxy both speculative participation and the ability of arbitrage capital to enter or exit the stock. The turnover spread is treated as a trading-friction characteristic rather than a pure measure of sentiment sensitivity.

4.6.11 Idiosyncratic Volatility

$IVOL_{i,t}^{FF3}$ is estimated from daily returns within each security-month. The residual-volatility measure follows the logic of the Fama-French three-factor model, where excess stock returns are decomposed into exposure to the market excess return, a size factor, and a book-to-market factor (**?**). The fitted component captures return variation associated with common market, size, and value factors. The residual component captures firm-specific return variation left after those factor exposures have been removed.

Daily excess returns are regressed on the Swedish daily market, size, and value factors,

$$r_{i,d} - RF_d = \alpha_{i,t} + \beta_{i,t} MKT\_RF\_D_d + s_{i,t} SMB\_D_d + h_{i,t} HML\_D_d + \varepsilon_{i,d}. \tag{2}$$

The monthly idiosyncratic-volatility characteristic is the standard deviation of the residuals $\varepsilon_{i,d}$. High $IVOL_{i,t}^{FF3}$ firms form the sentiment-prone side of the sort, and the spread is coded as high idiosyncratic volatility minus low idiosyncratic volatility.

? show that idiosyncratic volatility is a limits-to-arbitrage variable, not only a risk descriptor. Their argument is that arbitrageurs cannot diversify away the position-level idiosyncratic risk that arises when correcting stock-specific mispricing. They show that the IVOL-return relation depends on mispricing direction. Among overpriced stocks, high-IVOL firms earn much lower subsequent benchmark-adjusted returns than low-IVOL firms. Among underpriced stocks, the positive high-IVOL effect is weaker. The asymmetry is consistent with short-sale constraints because correcting overpricing requires shorting, while correcting underpricing can be done by buying.

? also show that investor sentiment strengthens the IVOL effect in the predicted direction. After high sentiment, the negative relation between IVOL and future returns becomes stronger among overpriced stocks. The finding makes $IVOL_{i,t}^{FF3}$ directly relevant for this thesis. The sentiment index measures the market-wide state, while $IVOL_{i,t}^{FF3}$ identifies the part of the Swedish cross-section where sentiment-driven overpricing should be most difficult to arbitrage away. The available Swedish data do not observe short interest, securities lending constraints, or stock-level institutional ownership. Idiosyncratic volatility is an indirect proxy for arbitrage risk rather than a direct measure of short-sale constraints.

4.7 Winsorization, Decile Assignment, and Portfolio Sorting Variables

Continuous characteristic variables are winsorized before portfolio assignment so that extreme observations do not determine the cross-sectional breakpoints. Market equity, total risk, book-to-market, profitability scaled by book equity, dividend scaled by book equity, asset tangibility, sales growth, illiquidity, turnover, and idiosyncratic volatility are winsorized within month at the 1st and 99th percentiles. Age is winsorized within fiscal year because it changes mechanically with calendar time rather than with monthly market conditions. The winsorized values remain economically interpretable, but the portfolio sorts are less sensitive to isolated data errors, stale prices, and extreme accounting ratios.

Decile assignments use Swedish observations as the reference universe in each formation period. For each month, local breakpoints are computed from eligible Swedish common-equity observations and securities are assigned to deciles from 1 to 10. The lowest decile contains the lowest values of the sorting characteristic and the highest decile contains the highest values. The local construction differs from a direct U.S. replication using NYSE breakpoints, but it matches the empirical question of this thesis. The analysis tests whether sentiment predicts relative returns inside the Swedish cross-section, so the relevant benchmark is the Swedish distribution of size, risk, tangibility, sales growth, turnover, and idiosyncratic volatility.

Zero and non-positive accounting values carry economic meaning for earnings, dividends, and tangible assets. Non-positive earnings identify unprofitable firms, zero dividends identify non-dividend payers, and zero tangible assets identify firms with little observable asset backing. Binary characteristics are separated from continuous decile variables. Non-dividend-payer status and unprofitable status enter as binary portfolio sorts, while size, risk, age, profitability, tangibility, sales growth, illiquidity, turnover, and idiosyncratic volatility enter through decile assignments. For diagnostic accounting ratios, observations with non-positive earnings, zero dividends, or zero tangible assets are assigned to a separate zero category before the positive observations are ranked.

Portfolio formation uses one-month lagged sorting signals. At month t, each security is assigned to a portfolio using the characteristic value observed at $t-1$, and the return measured over the following holding period is then attached to that portfolio. Lagged assignment aligns the portfolio sort with information available before the return is realized and prevents the realized return month from determining the characteristic assignment.

The main empirical spreads are coded in the direction implied by the sentiment hypothesis. The sentiment-prone leg is formed from the side of the characteristic distribution that Baker and Wurgler identify as harder to value or harder to arbitrage. The comparison leg is formed from the more stable or less sentiment-sensitive side. For continuous characteristics, the directional spread uses the relevant extreme deciles: low-minus-high spreads are $P01 - P10$, while high-minus-low spreads are $P10 - P01$. Binary characteristics are unchanged because they naturally form two groups. Small firms are compared with large firms, high-risk firms with low-risk firms, high-idiosyncratic-volatility firms with low-idiosyncratic-volatility firms, low-profitability firms with high-profitability firms, low-tangibility firms with high-tangibility firms, high-growth firms with low-growth firms, unprofitable firms with profitable firms, non-dividend payers with dividend payers, illiquid firms with liquid firms, and low-turnover firms with high-turnover firms. With this coding, a negative coefficient on lagged sentiment means that the sentiment-prone leg earns lower future returns relative to the comparison leg after high sentiment.

4.8 Composite Sentiment Index

Investor sentiment is estimated as a latent market-wide component extracted from several noisy proxy variables. The top-down approach in **?** and **?** measures sentiment through common variation in proxies related to survey confidence, trading activity, IPO markets, financing choices, and the relative valuation of dividend-paying firms. **?** provide a related factor-extraction approach for combining sentiment measures. The main orthogonalized index is $SENT^{\perp}$. The raw comparison index is $SENT$.

4.8.1 Survey Confidence

ESI_t and CCI_t enter because survey confidence can contain direct information about optimism. **?** compare changes in Michigan consumer confidence with UBS and Gallup investor optimism, a survey that asks investors about expected stock-market performance over the next year. The strong correlation between the two survey series supports the use of confidence indicators as sentiment proxies. The Swedish index combines these survey measures with market-state variables available at monthly frequency.

4.8.2 Trading Activity and Financing Composition

$TURN_t$ captures the liquidity and trading-activity channel. Behavioral models link trading to heterogeneous beliefs and noise-trader demand, because optimistic investors can trade aggressively when beliefs diverge and arbitrage is limited (**??**). **?** argue that market liquidity can proxy for sentiment when short-sale constraints make pessimistic views harder to trade on than optimistic views. High turnover is interpreted as stronger optimistic participation in the market. **?** show that the equity share in new issues predicts low subsequent aggregate returns, which motivates the use of financing composition as a sentiment-related proxy. The Swedish ED_RATIO_t variable is an adaptation of this issuance-composition channel.

4.8.3 IPO Market Conditions

$NIPO_t$ and $RIPO_t$ capture hot-issue-market conditions. Baker and Wurgler use IPO volume and first-day IPO returns as sentiment proxies, drawing on the IPO-cycle literature in which IPO activity and initial returns move with market conditions (**???**). **?** and **?** show that IPO returns also contain sequential-learning and information-spillover effects, so the IPO variables are interpreted as market-state proxies with sentiment, valuation news, and information-production content.

IPO counts and IPO returns have different missing-data meanings. $NIPO_t$ is measured on the full monthly paper-sample calendar and equals zero in months with no IPO activity. Raw $RIPO_t$ is defined only when at least one first-day IPO return is observed, and RIPO_OBSERVED_FLAG records whether a monthly IPO-return observation contributes to the raw series. Standardized $RIPO_t$ is neutral-imputed only after standardization when PCA requires a balanced matrix.

4.8.4 Dividend Premium

$DIVP_t$ follows the dividend-premium channel in **?**. It is constructed as the log difference between value-weighted market-to-book ratios of regular dividend payers and non-payers. Dividend payers are typically larger, more profitable, and less growth-oriented firms (**?**). A lower dividend premium is consistent with stronger relative demand for speculative non-payers (**??**).

4.8.5 PCA Construction Trace

PCA construction follows the staged procedure used by **?**, applied to this thesis proxy set with the added dividend-premium channel. The first stage estimates a preliminary PCA on current and one-month-lagged versions of each proxy. The preliminary score determines the timing of each proxy family. The selected proxies are then used to construct the raw comparison index, $SENT$, and the macro-residualized index, $SENT^{\perp}$.

Retaining components until cumulative explained variance exceeds 85 percent is a methodological deviation from **?**, who construct their sentiment index from the first principal component after selecting proxy timing. The 85 percent retention rule follows the CICSI construction in **?**, where several sentiment proxies are combined through a cumulative-variance PCA threshold for the Chinese stock market. Later Chinese sentiment research applies the same standard when constructing composite sentiment indices, including the investor sentiment measures in **?**. The same 85 percent threshold is used in the preliminary, raw selected-proxy, and macro-residualized PCA steps in this thesis.

The preliminary PCA uses fourteen candidate variables, consisting of current and lagged values of the seven proxy families. Retained components explain 86.0 percent of standardized proxy variance. **??** reports the retained components as construction evidence for timing selection and component interpretation. The first component is mainly a financing and dividend-valuation component, while the second component captures trading activity. Survey confidence and IPO-market variables enter through later components. The full eigenvector matrix is reported in **??**.

Table 4: Preliminary PCA component interpretation for lead-lag treatment

Component	Variance	Cumulative	Largest absolute loadings	Interpretation
Comp1	0.255	0.255	ED_RATIO_{t-1} 0.510, ED_RATIO_t 0.509, $DIVP_t$ 0.453	Financing composition and dividend valuation
Comp2	0.186	0.441	$TURN_t$ 0.523, $TURN_{t-1}$ 0.506, $NIPO_{t-1}$ -0.317	Trading activity and IPO-market state
Comp3	0.136	0.577	ESI_t 0.482, CCI_{t-1} 0.473, CCI_t 0.455	Survey confidence
Comp4	0.106	0.683	$NIPO_t$ 0.604, $NIPO_{t-1}$ 0.540, ESI_{t-1} -0.285	IPO volume timing
Comp5	0.098	0.782	CCI_t -0.481, ESI_t -0.478, ESI_{t-1} 0.471	Survey timing
Comp6	0.041	0.822	$RIPO_t$ 0.555, CCI_{t-1} -0.445, ESI_{t-1} 0.421	IPO-return and survey residual variation
Comp7	0.037	0.860	$RIPO_{t-1}$ 0.600, $NIPO_t$ 0.360, $NIPO_{t-1}$ -0.354	IPO timing residual variation

Note. Largest loadings are selected by absolute value within each retained component. Signs are retained because they show component orientation. The 85% retention threshold follows CICSI-style PCA construction and differs from the first-component construction in **?**.

For each proxy candidate j, the code converts the retained component eigenvectors into a score coefficient,

$$w_j = \sum_{k=1}^{K} \ell_{jk} \frac{\lambda_k}{\sum_m \lambda_m}, \tag{3}$$

where ℓ_{jk} is the loading of variable j on retained component k, λ_k is the eigenvalue of component k, and K is the number of retained components. The preliminary index is the weighted sum of standardized proxy candidates, $\widehat{SENT}_t^{pre} = \sum_j w_j z_{j,t}$. Its role is timing selection.

Lead-lag selection compares the correlation between each current or lagged proxy and the preliminary PCA index. **??** selects ESI_{t-1}, CCI_{t-1}, $TURN_{t-1}$, $NIPO_t$, $RIPO_t$, ED_RATIO_{t-1}, and $DIVP_t$. The correlation gap shows how strong each timing decision is. The dividend-premium timing is almost indifferent, with a gap of 0.002 between current and lagged values. Survey confidence has a clearer lagged timing. $RIPO_t$ contributes weak timing information, with a selected correlation of 0.001.

Table 5: Lead-lag selection for sentiment proxies

Proxy	Current corr.	Lag corr.	Corr. gap	Selected timing
ESI	0.003	**0.105**	0.102	Lagged
CCI	0.058	**0.125**	0.067	Lagged
TURN	0.419	**0.448**	0.028	Lagged
NIPO	**0.167**	0.146	0.020	Current
RIPO	**0.001**	-0.040	0.041	Current
ED_RATIO	0.607	**0.614**	0.008	Lagged
DIVP	**0.894**	0.892	0.002	Current

Note. The correlation gap is the selected correlation minus the alternative timing correlation. The selected timing is the version with the higher correlation with the preliminary PCA index.

After timing is fixed, the selected raw proxies enter a second PCA. Five retained components explain 89.9 percent of standardized variance, with the first component explaining 25.9 percent. **??** shows that the raw index is most closely tied to survey confidence and dividend valuation. ESI_{t-1} and CCI_{t-1} have the largest raw score coefficients and the highest correlations with $SENT_t$. Before macro adjustment, the raw index is a confidence-and-valuation composite.

Table 6: Raw sentiment index composition

Proxy	Score coef. in $SENT_t$	Corr. with $SENT_t$	Corr. rank
ESI_{t-1}	0.262	0.699	1
CCI_{t-1}	0.203	0.668	2
$TURN_{t-1}$	0.121	0.166	6
$NIPO_t$	0.101	0.226	5
$RIPO_t$	0.058	0.093	7
ED_RATIO_{t-1}	0.122	0.445	4
$DIVP_t$	0.179	0.534	3

Note. Score coefficients are summed weighted component coefficients from the selected-proxy PCA. The retained raw components explain 89.9% of standardized selected-proxy variance.

$SENT^{\perp}$ applies the same selected proxy timing after macro residualization. Each selected proxy is regressed on normalized CPI_t, PPI_t, IP_t, and EM_t, and PCA is applied to the standardized residuals. Raw $RIPO_t$ remains missing outside months with observed IPO first-day returns. Standardized $RIPO_t$ and residualized $rRIPO_t$ receive neutral imputation after standardization when PCA requires a balanced matrix.

The macro-residualized PCA has a stronger first component than the raw selected-proxy PCA. The first component explains 31.2 percent of residualized proxy variance, and five retained components explain 87.4 percent. **??** reports the final composition of $SENT_t^{\perp}$. The largest final coefficients belong to turnover, the dividend premium, and the equity-to-debt ratio. The same three proxies also have the highest correlations with the final index.

Table 7: Macro-orthogonalized sentiment index composition

Residualized proxy	Score coef. in $SENT_t^{\perp}$	Corr. with $SENT_t^{\perp}$	Corr. rank
$rESI_{t-1}$	0.115	0.127	6
$rCCI_{t-1}$	0.016	0.044	7
$rTURN_{t-1}$	0.250	0.756	2
$rNIPO_t$	0.150	0.221	4
$rRIPO_t$	0.064	0.138	5
rED_RATIO_{t-1}	0.199	0.754	3
$rDIVP_t$	0.224	0.773	1

Note. Score coefficients are summed weighted component coefficients from the residualized selected-proxy PCA. The retained macro-orthogonalized components explain 87.4% of standardized residualized proxy variance.

?? compares the raw and macro-orthogonalized indices proxy by proxy. Macro adjustment shifts index influence away from survey confidence and toward market-based residual variation. The correlation between economic sentiment and the index falls from 0.699 to 0.127, while the correlation between turnover and the index rises from 0.166 to 0.756. The dividend premium and equity-to-debt ratio also become more central after residualization. The correlation between $SENT_t$ and $SENT_t^{\perp}$ is 0.548, so $SENT^{\perp}$ preserves part of the raw common component while changing its composition.

Table 8: Raw and macro-orthogonalized index composition compared

Proxy family	Raw coef.	Raw corr.	Orth. coef.	Orth. corr.	Corr. change	Interpretation
ESI_{t-1}	0.262	0.699	0.115	0.127	-0.572	Survey influence falls
CCI_{t-1}	0.203	0.668	0.016	0.044	-0.624	Survey influence falls
$TURN_{t-1}$	0.121	0.166	0.250	0.756	0.590	Trading residual becomes central
$NIPO_t$	0.101	0.226	0.150	0.221	-0.005	IPO volume remains moderate
$RIPO_t$	0.058	0.093	0.064	0.138	0.045	IPO returns remain weak
ED_RATIO_{t-1}	0.122	0.445	0.199	0.754	0.309	Financing residual becomes central
$DIVP_t$	0.179	0.534	0.224	0.773	0.239	Dividend valuation becomes central

Note. Correlation change equals the macro-orthogonalized correlation minus the raw correlation. Orthogonalized coefficients omit the residual prefix to keep proxy families comparable across columns.

4.9 Macro-Adjusted Sentiment

The macro-adjusted sentiment path residualizes the sentiment proxies against Swedish macro-control variables before extracting the common component. The available macro controls are consumer prices, CPI_t, producer prices, PPI_t, industrial production, IP_t, and employment, EM_t. The controls are normalized and used in proxy-level residual regressions, after which PCA is applied to the residualized proxy series. The purpose is to separate the component of the sentiment proxy set that is mechanically related to contemporaneous macroeconomic conditions from the residual component more plausibly interpreted as sentiment.

Macro adjustment is an interpretation safeguard. The residualized index reduces business-cycle contamination and makes the remaining common component more defensible as a sentiment state rather than a direct macroeconomic state.

4.10 Risk-Free Rate and Factor Controls

4.10.1 Theoretical Role of Factor Controls

? estimate predictive regressions with market, size, value, and momentum controls to test whether sentiment-conditioned return patterns survive standard return-factor adjustment. Their controls combine the market excess return, the Fama-French size and value factors, and a momentum factor (**??**). The Swedish regressions follow the same logic. $RMRF_t$, SMB_t, HML_t, and UMD_t absorb broad market movements and common return components associated with firm size, book-to-market, and prior returns. The coefficient on lagged sentiment is then interpreted as predictive variation in the directional spread after conditioning on these factor returns.

Factor controls strengthen the empirical interpretation without turning the test into causal identification. A significant sentiment coefficient after factor adjustment is less likely to reflect ordinary exposure to the market, size, value, or momentum factors. It remains a predictive association between lagged sentiment and later return spreads.

4.10.2 Swedish Construction From Available Data

Baker and Wurgler use $RMRF$, SMB, HML, and UMD as conditioning variables in their multivariate predictive regressions (?). Their Table V first reports sentiment-only regressions and then re-estimates the same long-short portfolio regressions with the market excess return, the Fama-French size and value factors, and momentum included. The role of the four controls is to test whether the sentiment coefficient remains after absorbing standard return variation associated with broad market exposure, firm size, book-to-market, and prior returns. Their specification omits SMB when the dependent portfolio is the size spread and omits HML when the dependent portfolio is the book-to-market spread, because those controls would mechanically overlap with the dependent variable. The Swedish regressions follow the same factor-adjustment logic, while using locally constructed Swedish controls and the final set of sentiment-prone directional spreads.

The factor controls are constructed from the same Swedish data as the sentiment-conditioned portfolios. The inputs are the resolved monthly market panel, the common-equity screen, point-in-time fundamental snapshots, and Swedish interest-rate series. This keeps the control returns aligned with the sample, source-resolution rules, return units, and portfolio timing used in the main regressions.

Monthly factor returns are measured in percentage points. Eligible securities must pass the common-equity screen, have valid monthly returns, and have positive lagged market equity when value-weighted returns are computed. The factor audit table records breakpoints, constituent counts, lagged weight sums, equal-weighted leg returns, value-weighted leg returns, and thin-leg flags.

4.10.3 Market Excess Return

The Swedish 3-month interbank rate is converted from an annual percentage rate into the monthly risk-free proxy RF_t^{3M}. The value-weighted local market return, $R_t^{M,VW}$, is computed from Swedish common-equity returns using lagged market equity as the portfolio weight. The market excess return is

$$RMRF_t = R_t^{M,VW} - RF_t^{3M}. \tag{4}$$

The 10-year Swedish yield, Y_t^{10Y}, is retained as a macro-financial series. The regression risk-free leg uses RF_t^{3M}.

4.10.4 Size Factor

SMB_t is constructed from monthly Swedish size sorts. Eligible common equities are sorted into low, middle, and high size buckets using the 30th and 70th percentile breakpoints of lagged market equity. The factor return uses the two extreme legs and is computed as the value-weighted return on small firms minus the value-weighted return on big firms,

$$SMB_t = R_t^{Small,VW} - R_t^{Big,VW}. \tag{5}$$

The middle bucket is retained for audit and breakpoint structure, while the factor value is based on the low and high size legs.

4.10.5 Value Factor

HML_t is constructed from monthly Swedish book-to-market sorts. Book equity is taken from the point-in-time accounting snapshots, while market equity comes from the resolved monthly market panel. The book-to-market signal is lagged before sorting. Eligible common equities are assigned to low, middle, and high book-to-market buckets using 30th and 70th percentile breakpoints. The factor return is the value-weighted return on high book-to-market firms minus the value-weighted return on low book-to-market firms,

$$HML_t = R_t^{High\ BE/ME,VW} - R_t^{Low\ BE/ME,VW}. \tag{6}$$

4.10.6 Momentum Factor

UMD_t is constructed from monthly Swedish momentum sorts. The momentum signal compounds prior monthly returns while skipping the most recent month, following the convention that recent one-month reversals should be separated from medium-term momentum. Eligible common equities are sorted into loser, middle, and winner buckets using the 30th and 70th percentile breakpoints of the lagged momentum signal. The factor return is the value-weighted return on winner firms minus the value-weighted return on loser firms,

$$UMD_t = R_t^{Winner,VW} - R_t^{Loser,VW}. \tag{7}$$

4.10.7 Regression Use

The dependent return spread and the factor controls are measured over the same return horizon. For horizons of 1, 3, 6, and 12 months, forward factor windows match the compounded portfolio-return window from month t through month $t + h - 1$. The regression equations are specified in the regression-design subsection.

4.11 Idiosyncratic Volatility Extension

The IVOL extension estimates monthly idiosyncratic volatility from daily excess returns. The Fama-French three-factor model represents excess stock returns as a linear function of the market excess return, a size factor, and a value factor (?). The model works by estimating a stock's factor exposures and then treating the residual as return variation that is not explained by those common factors. The Swedish adaptation applies that structure at daily frequency within each security-month. It first constructs daily Swedish factors by converting the monthly risk-free proxy into daily rates and estimating daily market, size, and value factor returns. $IVOL_{i,t}^{FF3}$ is the monthly standard deviation of residuals from the daily three-factor regression. A minimum of 15 valid daily observations is required, and status flags identify months with missing returns, missing factor inputs, or insufficient observations.

$IVOL_{i,t}^{FF3}$ proxies for arbitrage risk in the sentiment-conditioned tests. The empirical results use this single residual-volatility measure to avoid reporting mechanically redundant IVOL evidence.

4.12 Portfolio Test Design

The portfolio tests translate the cross-sectional prediction in **?** into monthly Swedish return series. Each month, eligible securities are sorted using one-month lagged characteristics. Continuous variables are assigned to decile portfolios, while binary variables form two portfolios directly. Returns are measured after portfolio formation, so the dependent variables use information available before the holding-period return is realized. Equal-weighted portfolio returns are used because the theory concerns cross-sectional differences in sentiment sensitivity rather than the value-weighted performance of the largest Swedish firms.

The first empirical design follows the long-short portfolio logic in **?**. Baker and Wurgler form characteristic portfolios for size, age, volatility, profitability, dividend policy, asset tangibility, growth opportunities, and distress. Their central prediction is that sentiment should affect stocks most strongly when valuation is subjective and arbitrage is costly. In their U.S. tests, subsequent returns are relatively low after high sentiment for small, young, volatile, unprofitable, non-dividend-paying, and otherwise speculative firms. This thesis implements the same logic by coding each Swedish directional spread as the return on the sentiment-prone leg minus the return on the more sentiment-insensitive leg. A negative coefficient on lagged sentiment means that high sentiment predicts lower future returns for the Swedish stocks expected to be most sensitive to sentiment.

The Swedish spread design inspired by **?** is a broad cross-sectional test. It asks whether the relative return between two characteristic groups moves in the direction predicted by sentiment theory. For continuous characteristics, the dependent spread is formed from the sentiment-prone extreme decile minus the opposite extreme decile. If the sentiment-prone side is low values of the characteristic, the spread is $P01 - P10$. If the sentiment-prone side is high values, the spread is $P10 - P01$. Binary characteristics use non-dividend payers minus dividend payers or unprofitable firms minus profitable firms. The spread combines both sides of the comparison into one dependent variable and tests whether the cross-section tilts away from sentiment-prone firms after high sentiment. The leg decomposition then separates whether the effect comes from unusually weak returns on the sentiment-prone firms, unusually strong returns on the comparison firms, or both.

The second empirical design decomposes each directional spread into its two portfolio legs. For continuous characteristics, these legs are the same extreme deciles used in the spread construction, $P01$ and $P10$, with the ordering determined by the expected sentiment-prone side. For binary characteristics, the legs are the two binary groups. **?** combine market-wide sentiment with Miller's short-sale asymmetry. Their argument is that overpricing should be more common than underpricing when short-sale impediments prevent pessimistic investors from fully expressing negative views. In their anomaly tests, long-short anomaly profits are stronger after high sentiment, the short legs have lower subsequent returns after high sentiment, and the long legs show little relation to sentiment. The sentiment-prone Swedish leg is the closest analogue to the overpriced short leg in Stambaugh, Yu, and Yuan, even though the Swedish portfolios are characteristic spreads rather than anomaly trading strategies.

The two designs answer related but different questions. The spread test based on **?** asks whether Swedish sentiment predicts relative returns across characteristic-sorted portfolios. The Stambaugh-style leg test asks where that predictability comes from. Evidence is strongest when a characteristic has a negative spread coefficient and the sentiment-prone leg also has a negative coefficient. A spread driven mainly by the comparison leg still shows predictability, but it gives weaker support for the overpricing interpretation because the Stambaugh mechanism predicts that high sentiment should mainly depress future returns on the overpriced, difficult-to-arbitrage side.

4.13 Regression Design

4.13.1 Predictive Object

The empirical regressions estimate the predictive relations defined in the portfolio test design. The dependent variable is a characteristic-sorted portfolio return, either a sentiment-prone-minus-insensitive spread or one of its two component legs (**??**).

Ordinary least squares is used because the empirical object is a linear predictive relation between a lagged state variable and future portfolio returns. The coefficient on lagged sentiment has a direct quantitative interpretation in percentage points. It gives the expected change in the future return spread associated with a one-unit increase in sentiment, conditional on the included factor returns. The regression unit is a characteristic-horizon time series, with separate models estimated for each characteristic and return horizon.

4.13.2 Directional-Spread Regression

The primary spread test based on **?** uses the compounded future return on the sentiment-prone leg minus the compounded future return on the insensitive leg. For characteristic j and horizon h, the estimated model is

$$\begin{aligned} R^{P-S}_{j,t:t+h-1} = \alpha_j + \beta_j SENT^{\perp}_{t-1} + \gamma_j RMRF_{t:t+h-1} + \delta_j SMB_{t:t+h-1} \\ + \eta_j HML_{t:t+h-1} + \theta_j UMD_{t:t+h-1} + \varepsilon_{j,t+h}. \end{aligned} \tag{8}$$

$R^{P-S}_{j,t:t+h-1}$ is measured in percentage points and compounds monthly returns from month t through month $t+h-1$. $SENT^{\perp}_{t-1}$ is the lagged macro-orthogonalized sentiment index. The factor controls are compounded over the same future horizon as the dependent return. The estimated coefficient β_j is the conditional sentiment-predictability coefficient. A negative β_j means that high sentiment predicts lower future returns for the sentiment-prone side of characteristic j relative to the insensitive side.

For continuous characteristics, $R^{P-S}_{j,t:t+h-1}$ is the difference between the sentiment-prone extreme decile portfolio and the sentiment-insensitive extreme decile portfolio. Thus, low-minus-high characteristics use $P01 - P10$, while high-minus-low characteristics use $P10 - P01$. For binary characteristics, $R^{P-S}_{j,t:t+h-1}$ is the return on the sentiment-prone binary group minus the return on the safer binary group. A mechanical $P10 - P01$ diagnostic is still retained for audit purposes where the raw high-minus-low ordering is useful for comparison.

The same regression is also estimated with $SENT_{t-1}$ in place of $SENT^{\perp}_{t-1}$. That comparison isolates the empirical role of macro orthogonalization. A stronger coefficient for $SENT^{\perp}_{t-1}$ means that the macro-adjusted component of the index carries more return-predictive information than the raw proxy combination.

4.13.3 Leg-Decomposition Regression

The Stambaugh-style leg decomposition estimates the same predictive equation separately for the sentiment-prone leg and the insensitive leg. For each leg ℓ, the model is

$$\begin{aligned} R^{\ell}_{j,t:t+h-1} = \alpha_{j,\ell} + \beta_{j,\ell} SENT^{\perp}_{t-1} + \gamma_{j,\ell} RMRF_{t:t+h-1} + \delta_{j,\ell} SMB_{t:t+h-1} \\ + \eta_{j,\ell} HML_{t:t+h-1} + \theta_{j,\ell} UMD_{t:t+h-1} + \varepsilon_{j,\ell,t+h}. \end{aligned} \tag{9}$$

The leg regressions identify whether spread predictability comes from weak subsequent returns on the sentiment-prone side, strong subsequent returns on the insensitive side, or both. For continuous characteristics, the leg regressions use the same extreme $P01$ and $P10$ portfolios as the directional spread. This focuses the Stambaugh-style test on the firms most likely to be overpriced under the sentiment mechanism. The theoretically cleaner overpricing interpretation requires a negative coefficient on the sentiment-prone leg after high sentiment. A spread result driven mainly by the insensitive leg is still predictive evidence, but it is less directly aligned with the short-leg asymmetry in **?**.

4.13.4 Sentiment-Only and Factor-Adjusted Comparison

The factor-control robustness check estimates the predictive spread regression twice on the same complete-case sample. The sentiment-only specification is

$$R^{P-S}_{j,t:t+h-1} = \alpha_j + \beta_j SENT^{\perp}_{t-1} + \varepsilon_{j,t+h}. \tag{10}$$

The factor-adjusted specification adds $RMRF$, SMB, HML, and UMD. The comparison separates standalone explanatory power from conditional predictability. The sentiment-only R^2 shows how much variation in the directional spread is explained by lagged sentiment alone. The factor-adjusted R^2 shows how much variation is explained after standard return factors enter the model. The change in β_j shows whether the sentiment relation becomes weaker or stronger once common market, size, value, and momentum variation is absorbed.

4.13.5 Non-Overlapping Horizon Regressions

The non-overlapping robustness test re-estimates the directional-spread regression on calendar-offset subsamples. For horizon h, the sample is split into h subsamples according to the month index,

$$m(t) \bmod h = k, \qquad k = 0, 1, \ldots, h-1. \tag{11}$$

Each offset subsample contains return windows that are separated by the full return horizon. The model is otherwise the same factor-adjusted predictive OLS regression as the primary specification. The resulting median coefficient and sign-consistency statistics show whether the $SENT^{\perp}$ result survives when adjacent observations share less of the same future return window.

4.13.6 Sentiment-Innovation Regression

The sentiment-innovation test removes first-order persistence from $SENT^{\perp}$ before the predictive regression is estimated. The index is first modelled as

$$SENT_t^{\perp} = a + \rho SENT_{t-1}^{\perp} + u_t. \tag{12}$$

The residual u_t is the innovation in sentiment, measured as the component of current sentiment left after projecting the index on its own lag. The directional-spread regression is then estimated with u_{t-1} replacing $SENT_{t-1}^{\perp}$. This test asks whether unexpected changes in sentiment predict future cross-sectional return spreads. The $SENT^{\perp}$ level regression asks whether persistent high- and low-sentiment states predict future reversals.

4.13.7 Auxiliary Regressions

Two auxiliary OLS regressions enter before the final predictive tests. The macro-orthogonalized sentiment index residualizes each selected sentiment proxy against Swedish macroeconomic controls,

$$Z_{k,t} = a_k + b_{1k}CPI_t + b_{2k}PPI_t + b_{3k}IP_t + b_{4k}EM_t + e_{k,t}. \tag{13}$$

The residuals $e_{k,t}$ are standardized and passed into the PCA index construction. This regression step reduces business-cycle contamination in the sentiment proxies before the return-predictability tests are estimated.

Idiosyncratic volatility is estimated from daily stock-level excess returns within each security-month. The three-factor residual-volatility model is

$$r_{i,d} - RF_d = \alpha_{i,m} + \beta_{i,m}MKT\_RF_d + s_{i,m}SMB_d + h_{i,m}HML_d + \varepsilon_{i,d}. \tag{14}$$

The monthly characteristic $IVOL_{i,m}^{FF3}$ is the standard deviation of the fitted residuals $\varepsilon_{i,d}$. This OLS regression constructs a stock-level measure of arbitrage risk for the subsequent characteristic sorts.

4.13.8 HAC-Newey-West and Bootstrap t-statistics

Standard OLS t-statistics use a covariance estimator that treats regression errors as homoskedastic and serially uncorrelated. The predictive regressions in this thesis use monthly financial returns, persistent sentiment variables, and multi-month forward return windows. These features create two econometric problems. Residual variance can change over time, and residuals can be serially correlated because adjacent 3-, 6-, and 12-month return windows share many of the same monthly returns. The coefficient estimate remains an OLS estimate, but the conventional OLS standard error becomes a weak basis for inference when the error covariance matrix is misspecified.

Newey-West inference replaces the conventional OLS covariance matrix with a heteroskedasticity- and autocorrelation-consistent covariance estimator (**?**). **?** use Newey-West standard errors in their German sentiment regressions, where monthly sentiment and return relations are estimated in time series. **?** also report Newey-West standard errors and bootstrap inference in sentiment-return regressions, explicitly because the tests involve heteroskedasticity and serial correlation. The point estimate $\hat{\beta}$ is still the OLS coefficient. The difference is the estimated standard error used in the t-statistic,

$$t_{\beta}^{HAC} = \frac{\hat{\beta}}{\widehat{se}_{HAC}(\hat{\beta})}. \tag{15}$$

The HAC standard error is computed from autocovariances of the regression residuals with declining Bartlett weights. The lag length is at least $h-1$ for a horizon of h months, which directly addresses the mechanical overlap in the dependent return windows. Empirical sentiment and anomaly papers use robust time-series inference for the same reason. The return-predictability question concerns whether a coefficient survives serial dependence in monthly portfolio returns. Independent-error inference is too narrow for that question (**?**).

Baker and Wurgler report bootstrapped p-values in their long-short portfolio regressions because an autocorrelated sentiment index can create finite-sample bias when innovations in sentiment are correlated with innovations in portfolio returns (**?**). The bootstrap approach used in this thesis follows the same motivation. It treats persistence and time-series dependence as part of the sampling problem and reduces reliance on a single analytical covariance formula.

The moving-block bootstrap resamples blocks of monthly regression observations, re-estimates the OLS regression in each bootstrap sample, and computes the dispersion of the resulting coefficient estimates. The bootstrap t-statistic is

$$t_{\beta}^{Boot} = \frac{\hat{\beta}}{\widehat{se}_{Boot}(\hat{\beta})}. \tag{16}$$

The bootstrap standard error $\widehat{se}_{Boot}(\hat{\beta})$ is the standard deviation of the bootstrap coefficient estimates. This differs from the HAC t-statistic because the standard error is generated from repeated block-resampled regressions. HAC inference corrects the estimated covariance matrix in the original sample. Bootstrap inference asks whether the coefficient remains large relative to the empirical distribution generated by resampling dependent blocks of the data.

4.14 Diagnostics and Validation

Diagnostics are part of the empirical design. Market-source overlap diagnostics document the hybrid monthly source resolution. Duplicate-key checks protect the security-month and company-month panels. Variable-availability audits document characteristic coverage. Factor-leg audits record whether the Swedish factor portfolios have usable inputs. IVOL status flags show whether monthly residual-volatility estimates are available or blocked by insufficient inputs. Sentiment-index diagnostics record the selected lead-lag proxy set, PCA coefficients, and macro-orthogonalized proxy weights.

The diagnostics separate data-quality checks from unresolved limitations. Measurement error in a constructed panel can otherwise be mistaken for asset-pricing evidence.

4.15 Methodological Limitations and Deliberately Secondary Elements

Global, Nordic, and U.S. sentiment spillovers are relevant in the literature and belong to later extensions beyond the Sweden-first design. The empirical design is interpreted as return-predictability evidence. A causal interpretation would require a separate specification with a credible identification strategy.

5 Investigation, Analysis and Results

Directional spreads based on **?** provide the broad evidence, while the Stambaugh-style leg decomposition gives a stricter test of where predictability comes from. High sentiment predicts lower subsequent returns for sentiment-prone Swedish portfolios, especially total risk, idiosyncratic volatility, low tangibility, size, and dividend status. The leg evidence is more demanding and more mixed. Total risk, idiosyncratic volatility, and size give the cleanest joint support, while low tangibility is strong in spreads but partly driven by the insensitive leg. Sales growth is correctly signed but weak. Profitability, illiquidity, and turnover provide weak or conflicting evidence.

5.1 Sentiment Index and Portfolio Setup

The final sentiment index is $SENT_t^{\perp}$, a macro-orthogonalized Swedish PCA index that combines survey confidence, turnover, IPO activity, IPO first-day returns, the equity-to-debt proxy, and the dividend premium. The raw comparison index, $SENT_t$, uses the same proxy set before macro residualization. The raw and orthogonalized series have a correlation of 0.548, so macro adjustment changes the empirical sentiment state materially.

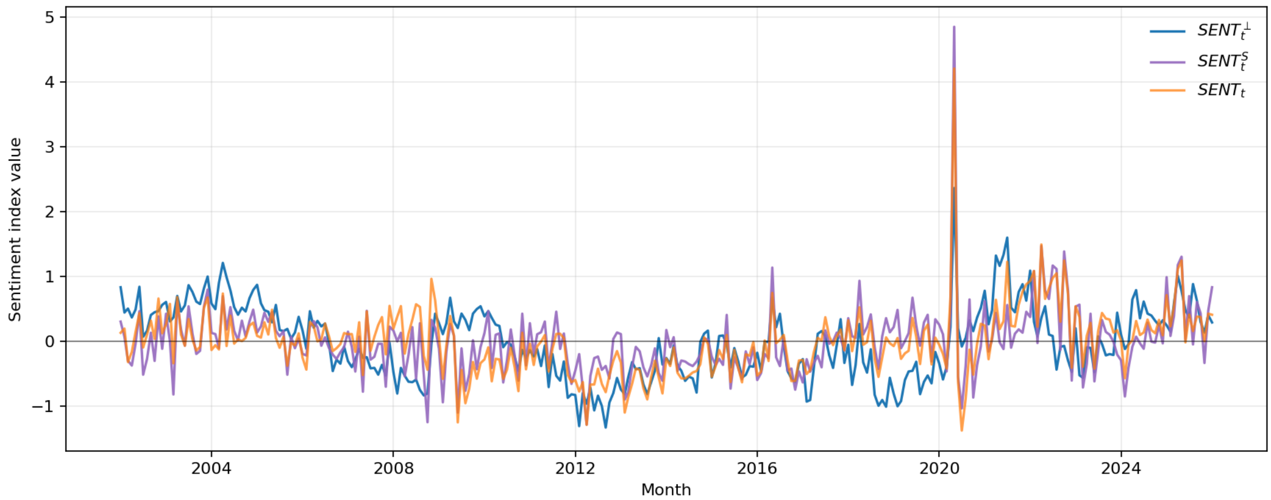

Figure 1: Raw and macro-orthogonalized Swedish dividend-premium sentiment indices

Note. The figure plots $SENT_t$ and $SENT_t^{\perp}$.

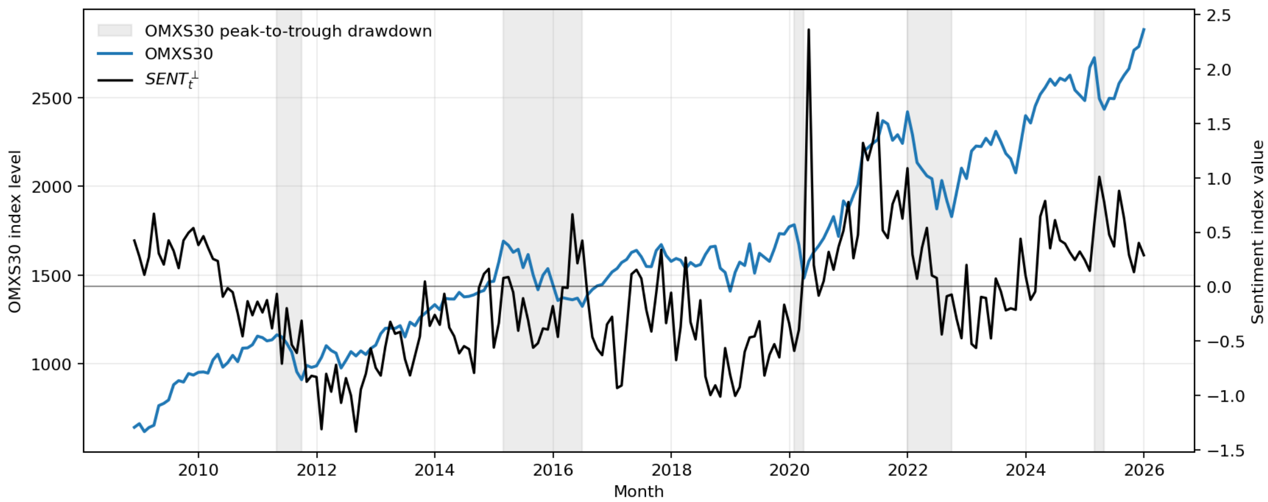

Figure 2: OMXS30 and macro-orthogonalized sentiment during major drawdowns

Note. The blue line is the OMXS30 adjusted index level on the left axis. The black line is $SENT_t^{\perp}$ on the right axis. Grey areas mark the five largest peak-to-trough OMXS30 drawdowns in the monthly OMXS30 series.

?? places $SENT_t^{\perp}$ against large-cap Swedish market stress. Elevated sentiment is followed by weak aggregate returns around the 2021–2022 market decline. Other drawdowns occur when $SENT_t^{\perp}$ is moderate, and some high-sentiment observations are followed by no immediate large OMXS30 decline. The aggregate index relation is weaker than the cross-sectional prediction tested below.

OMXS30 is a large-cap benchmark made up of liquid, established Swedish firms. The mechanism in **?** predicts larger sentiment effects in firms whose values are harder to estimate and harder to arbitrage, not necessarily in a blue-chip index dominated by larger firms with more durable cash-flow profiles. The sentiment index moves through important Swedish stress periods, but the return-predictability evidence comes from the characteristic-sorted cross-section.

Table 9: Sentiment-index summary statistics

Index	Obs.	Mean	Std. dev.	Min.	Median	Max.
$SENT$	289	-0.009	0.519	-1.379	-0.006	4.205
$SENT^{\perp}$	289	-0.014	0.558	-1.333	-0.043	2.361
$\Delta SENT^{\perp}$	288	-0.002	0.356	-2.163	-0.012	2.143

Note. The raw and orthogonalized index have a correlation of 0.548.

The final directional portfolio design uses eleven characteristics: ME, age, risk, $IVOL^{FF3}$, non-dividend-payer status, unprofitable status, $E+BE$, PPE/A, GS, $ILLIQ$, and $XTURN$. BE/ME remains a diagnostic value sort rather than a main directional sentiment-prone test.

Table 10: Directional portfolio definitions

Sort variable	Sentiment-prone leg	Directional spread
ME	Small firms	Low minus high
$risk$	High total volatility	High minus low
$IVOL^{FF3}$	High three-factor idiosyncratic volatility	High minus low
Age	Young firms	Low minus high
Non-dividend payer	Non-payers	Non-payers minus payers
Unprofitable	Non-positive earnings firms	Unprofitable minus profitable
$E + BE$	Low profitability	Low minus high
PPE/A	Low tangibility	Low minus high
GS	High sales growth	High minus low
Illiquidity	High Amihud illiquidity	High minus low
Turnover	Low share turnover	Low minus high

Note. For continuous characteristics, low minus high means $P01 - P10$; high minus low means $P10 - P01$. Binary characteristics use their two natural groups. BE/ME is retained only as a diagnostic value sort because the sentiment-prone side is not monotonic and the dividend-premium proxy is valuation-related.

The directional spread is always the return on the sentiment-prone side minus the return on the sentiment-insensitive side. A negative coefficient on lagged sentiment means that the sentiment-prone portfolio earns lower subsequent returns after high sentiment. Negative coefficients are the hypothesis-consistent sign in the spread tests based on **?**. The later leg-decomposition tables estimate the two component legs separately.

Table 11: Correlations of directional portfolio spreads

	ME	Age	Risk	Non-payer	Unprof.	PPE/A	GS	IVOL FF3	ILLIQ	XTURN	E+BE
ME	1.00										
Age	0.11	1.00									
Risk	0.51	-0.05	1.00								
Non-payer	0.70	0.16	0.61	1.00							
Unprof.	0.39	0.18	0.15	0.40	1.00						
PPE/A	0.26	0.02	0.27	0.22	0.13	1.00					
GS	-0.04	0.09	0.05	-0.06	-0.13	-0.09	1.00				
IVOL FF3	0.62	-0.04	0.81	0.65	0.20	0.20	0.03	1.00			
ILLIQ	0.81	0.14	0.45	0.66	0.32	0.19	-0.06	0.59	1.00		
XTURN	-0.28	0.21	-0.56	-0.47	-0.13	-0.22	0.07	-0.54	-0.07	1.00	
E+BE	0.17	0.02	-0.01	0.18	0.81	0.08	-0.12	0.02	0.11	-0.03	1.00

Note. The table reports pairwise correlations of monthly directional spread returns. Pairwise monthly observations range from 262 to 311.

Several spreads load on the same underlying part of the Swedish cross-section. The IVOL spread is strongly related to total risk, and size is also closely related to illiquidity and dividend status. The sales-growth spread is only weakly correlated with the main risk and size spreads. The results describe a connected cluster of small, risky, illiquid, non-dividend-paying firms plus a more separate growth-opportunity characteristic, rather than eleven independent tests.

5.2 Baker-Wurgler Style Directional Spreads

5.2.1 High- and Low-Sentiment Portfolio Evidence

Average one-month spreads fall after high sentiment for total risk, three-factor IVOL, low tangibility, young firms, sales growth, and low turnover. The largest high-minus-low differences are observed for total risk, idiosyncratic volatility, and low tangibility. Non-dividend-payers, unprofitable firms, small firms, low profitability firms, and illiquid firms show weak or opposite simple high-low patterns in this descriptive split.

Table 12: Average one-month directional spreads after low and high sentiment

Characteristic	Low sentiment	High sentiment	High minus low
Total risk	-0.55	-1.57	-1.02
IVOL (FF3)	-1.06	-2.13	-1.07
Low tangibility	0.86	-0.04	-0.90
Non-dividend payer	-0.95	-0.72	0.23
Unprofitable	-1.43	-1.13	0.30
Small firms	0.40	0.45	0.05
Young firms	0.15	-0.30	-0.45
Low profitability	-1.56	-1.16	0.40
Sales growth	0.06	-0.20	-0.25
Illiquidity	-0.25	-0.01	0.24
Low turnover	0.71	0.15	-0.56

Note. Returns are monthly percentage points. Sentiment states are split at the median of lagged $SENT^{\perp}$.

5.2.2 Factor-Adjusted Predictive Regressions

The factor-adjusted regressions provide stronger evidence than the simple high-low split. The orthogonalized index produces more theory-consistent results than the raw index at every horizon. At the 12-month horizon, the average coefficient is -7.89 for $SENT_t^{\perp}$ and -4.00 for $SENT_t$. The share of characteristics with a theory-consistent 5% result is 0.64 for the orthogonalized index at 12 months, compared with 0.09 for the raw index.

Table 13: Predictive regression summary by return horizon

Horizon	Index	Mean coef.	Mean HAC t	Expected t	Bootstrap t	5% share
1	Orthogonalized	-0.97	-1.62	1.62	1.62	0.36
1	Raw	-0.59	-0.94	0.94	0.86	0.27
3	Orthogonalized	-2.04	-1.46	1.46	1.45	0.36
3	Raw	-1.03	-0.63	0.63	0.61	0.18
6	Orthogonalized	-4.37	-1.93	1.93	1.91	0.45
6	Raw	-2.63	-1.17	1.17	1.11	0.09
12	Orthogonalized	-7.89	-2.22	2.22	2.08	0.64
12	Raw	-4.00	-1.06	1.06	1.00	0.09

Note. The expected-signed t-statistic multiplies the coefficient t-statistic by the theoretically expected sign. Positive values support the sentiment-reversal hypothesis.

Each coefficient measures the percentage-point change in a future directional spread associated with a one-unit increase in lagged sentiment. Every spread is defined as sentiment-prone minus sentiment-insensitive, so a negative coefficient means that high sentiment predicts lower future returns for the sentiment-prone leg relative to the insensitive leg. The raw index columns report d and the HAC t-statistic. The orthogonalized index columns report d, the HAC t-statistic, and the moving-block bootstrap t-statistic. Stars on coefficients mark HAC significance at the 10, 5, and 1 percent levels.

Table 14: Predictive Regressions for Directional Sentiment-Prone Spreads: 1-Month Horizon

Variable	Spread	Raw index		Macro-orthogonalized index		
		Coef.	HAC t	Coef.	HAC t	Boot. t
Panel A. Size, Age, and Risk						
ME	Small minus big	-0.92	-1.17	-1.34*	-1.90	-1.92
Age	Young minus old	-0.40	-0.91	-0.38	-0.80	-0.83
σ	High minus low	-0.79	-1.19	-2.60***	-3.50	-3.51
$IVOL^{FF3}$	High minus low	-0.68	-0.85	-2.24***	-2.69	-2.69
Panel B. Profitability and Dividend Policy						
$E+BE$	Low minus high	0.28	0.58	-0.44	-0.64	-0.62
$Unprofitable$	>0 minus ≤ 0	-0.03	-0.08	-0.54	-0.96	-0.98
$Nonpayer$	Nonpayer minus payer	0.07	0.27	-0.63**	-2.24	-2.26
Panel C. Tangibility and Growth Opportunities						
PPE/A	Low minus high	-1.37**	-2.41	-1.27***	-2.99	-2.94
GS	High minus low	-1.10**	-2.18	-0.80	-1.33	-1.28
Panel D. Trading Frictions						
$ILLIQ$	High minus low	0.10	0.20	-0.30	-0.67	-0.66
$XTURN$	Low minus high	-1.68**	-2.56	-0.11	-0.16	-0.16

Note. Each row reports the coefficient on lagged sentiment in a factor-adjusted predictive regression of the directional spread return on sentiment and contemporaneous factor returns. For continuous characteristics, directional spreads use the sentiment-prone extreme decile minus the opposite extreme decile; binary spreads use the two natural groups. Negative coefficients are theory-consistent: high sentiment predicts lower subsequent returns for the sentiment-prone side. Raw index refers to $SENT$; macro-orthogonalized index refers to $SENT^{\perp}$. Coefficient stars mark HAC significance at 10%, 5%, and 1%.

Raw and Orthogonalized Sentiment **?** report raw and orthogonalized sentiment results side by

Table 15: Predictive Regressions for Directional Sentiment-Prone Spreads: 3-Month Horizon

Variable	Spread	Raw index		Macro-orthogonalized index		
		Coef.	HAC t	Coef.	HAC t	Boot. t
Panel A. Size, Age, and Risk						
ME	Small minus big	-0.52	-0.34	-2.44	-1.38	-1.39
Age	Young minus old	-1.53	-1.35	-1.06	-0.79	-0.74
σ	High minus low	-1.30	-0.81	-6.49***	-3.56	-3.48
$IVOL^{FF3}$	High minus low	-1.53	-0.85	-5.60***	-2.97	-3.01
Panel B. Profitability and Dividend Policy						
$E + BE$	Low minus high	1.06	1.25	0.43	0.34	0.32
$Unprofitable$	> 0 minus ≤ 0	0.41	0.75	-1.11	-1.36	-1.36
$Nonpayer$	Nonpayer minus payer	0.19	0.32	-1.88**	-2.46	-2.48
Panel C. Tangibility and Growth Opportunities						
PPE/A	Low minus high	-3.47***	-2.85	-4.46***	-3.86	-3.79
GS	High minus low	-1.88	-1.58	-0.88	-0.64	-0.63
Panel D. Trading Frictions						
$ILLIQ$	High minus low	0.72	0.85	-0.03	-0.03	-0.03
$XTURN$	Low minus high	-3.48**	-2.30	1.04	0.62	0.63

Note. Each row reports the coefficient on lagged sentiment in a factor-adjusted predictive regression of the directional spread return on sentiment and contemporaneous factor returns. For continuous characteristics, directional spreads use the sentiment-prone extreme decile minus the opposite extreme decile; binary spreads use the two natural groups. Negative coefficients are theory-consistent: high sentiment predicts lower subsequent returns for the sentiment-prone side. Raw index refers to $SENT$; macro-orthogonalized index refers to $SENT^{\perp}$. Coefficient stars mark HAC significance at 10%, 5%, and 1%.

Table 16: Predictive Regressions for Directional Sentiment-Prone Spreads: 6-Month Horizon

Variable	Spread	Raw index		Macro-orthogonalized index		
		Coef.	HAC t	Coef.	HAC t	Boot. t
Panel A. Size, Age, and Risk						
ME	Small minus big	-3.82	-1.41	-6.55**	-2.39	-2.39
Age	Young minus old	-3.28*	-1.87	-2.86	-1.23	-1.22
σ	High minus low	-4.22*	-1.65	-12.08***	-4.30	-4.25
$IVOL^{FF3}$	High minus low	-3.82	-1.36	-11.74***	-4.05	-4.02
Panel B. Profitability and Dividend Policy						
$E + BE$	Low minus high	0.98	0.63	0.41	0.18	0.18
$Unprofitable$	> 0 minus ≤ 0	0.04	0.04	-2.50*	-1.84	-1.76
$Nonpayer$	Nonpayer minus payer	-0.14	-0.14	-3.58***	-2.88	-2.91
Panel C. Tangibility and Growth Opportunities						
PPE/A	Low minus high	-6.02***	-3.40	-8.34***	-4.21	-4.11
GS	High minus low	-3.70*	-1.77	-1.41	-0.55	-0.54
Panel D. Trading Frictions						
$ILLIQ$	High minus low	-0.20	-0.15	-1.32	-0.65	-0.64
$XTURN$	Low minus high	-4.77*	-1.76	1.91	0.67	0.69

Note. Each row reports the coefficient on lagged sentiment in a factor-adjusted predictive regression of the directional spread return on sentiment and contemporaneous factor returns. For continuous characteristics, directional spreads use the sentiment-prone extreme decile minus the opposite extreme decile; binary spreads use the two natural groups. Negative coefficients are theory-consistent: high sentiment predicts lower subsequent returns for the sentiment-prone side. Raw index refers to $SENT$; macro-orthogonalized index refers to $SENT^{\perp}$. Coefficient stars mark HAC significance at 10%, 5%, and 1%.

Table 17: Predictive Regressions for Directional Sentiment-Prone Spreads: 12-Month Horizon

Variable	Spread	Raw index		Macro-orthogonalized index		
		Coef.	HAC t	Coef.	HAC t	Boot. t
Panel A. Size, Age, and Risk						
ME	Small minus big	-9.14	-1.61	-14.04***	-3.15	-2.84
Age	Young minus old	-6.33*	-1.82	-8.37**	-2.29	-2.11
σ	High minus low	-5.45	-1.09	-19.65***	-4.43	-4.18
$IVOL^{FF3}$	High minus low	-5.15	-1.07	-18.42***	-3.75	-3.76
Panel B. Profitability and Dividend Policy						
$E+BE$	Low minus high	2.08	0.92	0.08	0.02	0.02
$Unprofitable$	>0 minus ≤ 0	-0.99	-0.89	-4.86***	-3.04	-2.90
$Nonpayer$	Nonpayer minus payer	-0.66	-0.41	-4.96***	-2.80	-2.59
Panel C. Tangibility and Growth Opportunities						
PPE/A	Low minus high	-11.44***	-3.19	-15.27***	-4.83	-4.37
GS	High minus low	-3.88	-1.37	-3.21	-0.61	-0.58
Panel D. Trading Frictions						
$ILLIQ$	High minus low	-0.13	-0.05	-2.58	-0.72	-0.69
$XTURN$	Low minus high	-2.85	-1.10	4.46	1.23	1.14

Note. Each row reports the coefficient on lagged sentiment in a factor-adjusted predictive regression of the directional spread return on sentiment and contemporaneous factor returns. For continuous characteristics, directional spreads use the sentiment-prone extreme decile minus the opposite extreme decile; binary spreads use the two natural groups. Negative coefficients are theory-consistent: high sentiment predicts lower subsequent returns for the sentiment-prone side. Raw index refers to $SENT$; macro-orthogonalized index refers to $SENT^{\perp}$. Coefficient stars mark HAC significance at 10%, 5%, and 1%.

side. Their Table V shows that the two sentiment measures generally preserve the same empirical message for the central speculative-firm spreads. Orthogonalization changes magnitudes, but the raw index still predicts many of the same future return reversals. The Swedish evidence is different. Across the eleven directional spreads, $SENT$ has weaker average expected-signed HAC statistics than $SENT^{\perp}$ at every horizon. At the 12-month horizon, the mean coefficient changes from -4.00 for $SENT$ to -7.89 for $SENT^{\perp}$. The mean expected-signed HAC statistic changes from 1.06 to 2.22, and the share of theory-consistent 5% HAC results rises from 0.09 to 0.64.

The characteristic-level results show where the separation arises. $SENT$ gives strong evidence for low tangibility and some short-horizon evidence for low turnover and sales growth. It gives much weaker evidence for the characteristics that carry the clearest interpretation in **?**. At 12 months, the raw total-risk and $IVOL^{FF3}$ coefficients are negative but statistically weak, with HAC p-values of 0.275 and 0.285. The raw non-dividend-payer and unprofitable-firm coefficients are also weak, with HAC p-values of 0.685 and 0.374. The same four spreads under $SENT^{\perp}$ have HAC p-values of 0.000, 0.000, 0.005, and 0.002. The Swedish reversal evidence comes mainly from the macro-orthogonalized component rather than from the raw common component.

The PCA diagnostics give a plausible explanation. $SENT$ correlates most strongly with lagged economic sentiment and lagged consumer confidence, with correlations of 0.699 and 0.668. The raw index also correlates with the dividend premium and the equity-debt ratio, but turnover, IPO volume, and IPO returns contribute less to the raw common component. After macro residualization, the index composition changes sharply. $SENT^{\perp}$ correlates most strongly with residualized dividend premium, residualized turnover, and residualized equity-debt ratio, with correlations of 0.773, 0.756, and 0.754. These residualized market-based proxies are closer to the channels through which sentiment should predict cross-sectional reversals in hard-to-value and hard-to-arbitrage firms.

The raw Swedish index appears to combine investor optimism with broad confidence-cycle variation. The low correlation of 0.548 between $SENT$ and $SENT^{\perp}$ shows that macro orthogonalization changes the empirical state variable being tested. The results provide conditional support for the sentiment mechanism in **?** in the residual market-state component, while the raw Swedish index has limited ability to reproduce their cross-sectional pattern on its own.

Risk gives the clearest separation between the raw and orthogonalized indices. The raw total-risk coefficient is negative at all horizons, but its 12-month HAC p-value is 0.275. The orthogonalized 12-month coefficient is -19.65, with a HAC p-value of 0.000 and a bootstrap p-value of 0.000. The same pattern appears for $IVOL^{FF3}$. The raw 12-month coefficient is -5.15 with a HAC p-value of 0.285, while the orthogonalized coefficient is -18.42 with a HAC p-value of 0.000 and a bootstrap p-value of 0.000. A one-unit increase in lagged $SENT^{\perp}$ predicts substantially lower future returns for high-risk and high-IVOL firms relative to low-risk and low-IVOL firms. The risk and IVOL results most closely match **?**, where sentiment effects are strongest among firms with subjective valuations and stronger limits to arbitrage.

Size strengthens with the return horizon. The raw small-firm coefficient is negative at every horizon, but the HAC p-values stay above conventional 5% levels. The orthogonalized coefficient is -6.55 at 6 months with a HAC p-value of 0.017 and a bootstrap p-value of 0.022. At 12 months, it reaches -14.04 with a HAC p-value of 0.002 and a bootstrap p-value of 0.020. The pattern supports the prediction in **?** that small firms are more sentiment sensitive than large firms. Age is weaker. The orthogonalized 12-month coefficient is -8.37 with a HAC p-value of 0.022, but the bootstrap p-value is 0.086. The young-firm result has the expected sign and HAC support at 12 months, while the bootstrap evidence is closer to 10% significance than 5% significance.

Dividend policy also depends on the index definition. The raw non-dividend-payer coefficients are small and have weak HAC evidence. The orthogonalized coefficients are negative and statistically supported in every horizon table. At 12 months, the coefficient is -4.96 with a HAC p-value of 0.005 and a bootstrap p-value of 0.022. $SENT^{\perp}$ predicts lower future returns for non-dividend payers relative to dividend payers after high sentiment. The result matches the argument in **?** that firms without dividends are more sentiment sensitive because their valuation depends more on distant and uncertain cash flows. The Swedish interpretation remains cautious because $SENT^{\perp}$ contains the dividend premium as one of its proxies.

Profitability gives two different readings depending on the measure. The unprofitable-firm spread has the expected negative sign in the orthogonalized specification at every horizon and becomes statistically supported at 12 months, where the coefficient is -4.86, the HAC p-value is 0.002, and the bootstrap p-value is 0.014. The raw coefficient for the same 12-month spread is -0.99 with a HAC p-value of 0.374. Low profitability measured by $E + BE$ gives conflicting evidence. Its coefficients are positive at the 3-, 6-, and 12-month horizons for both raw and orthogonalized sentiment. The Swedish results support a profitability-related sentiment channel when profitability is measured as non-positive earnings, while the low $E + BE$ spread moves against the predicted reversal pattern.

Low tangibility is strong under both index definitions. The raw coefficient is negative and statistically supported under HAC inference at every horizon. At 12 months, the raw coefficient is -11.44 with a HAC p-value of 0.001. The orthogonalized 12-month coefficient is -15.27, with a HAC p-value of 0.000 and a bootstrap p-value of 0.000. A one-unit increase in lagged $SENT^{\perp}$ predicts substantially lower future returns for low-tangibility firms relative to high-tangibility firms. The evidence is consistent with this thesis mechanism because intangible firms are harder to value and easier to misprice. It is also stronger than the corresponding tangibility evidence in **?**, where tangibility is not among the clearest empirical patterns.

Sales growth has the expected sign but weak statistical support under $SENT^{\perp}$. The raw index gives a negative one-month coefficient and a weaker negative relation at longer horizons. The orthogonalized coefficient is also negative in every horizon, but the 12-month coefficient is -3.21 with a HAC p-value of 0.539 and a bootstrap p-value of 0.694. The Swedish sales-growth evidence adds the growth-opportunity characteristic from **?** to the final design, but it does not materially strengthen the headline result.

Trading frictions are the weakest part of the BW-style spread evidence. Illiquidity has unstable raw coefficients and weak orthogonalized coefficients. The orthogonalized coefficient has the expected negative sign at every horizon, but neither HAC nor bootstrap inference is close to conventional significance levels. Low turnover moves in opposite directions across the two indices. The raw index gives negative and statistically supported short-horizon coefficients, while the orthogonalized coefficient turns positive from 3 to 12 months. At the 12-month horizon, the orthogonalized coefficient is 4.46 with a HAC p-value of 0.220 and a bootstrap p-value of 0.294. $SENT^{\perp}$ predicts higher future returns for the low-turnover spread, which conflicts with the expected reversal logic.

The horizon-level pattern is supportive but depends strongly on macro orthogonalization. The raw index has expected-sign coefficients in several rows, especially low tangibility, but it produces much weaker HAC evidence across most characteristics. The orthogonalized index produces stronger and more consistent reversal evidence for risk, $IVOL^{FF3}$, low tangibility, dividend policy, and size at longer horizons. Age and unprofitable firms provide partial support. Sales growth, low profitability, illiquidity, and turnover weaken the overall interpretation. Compared with **?**, the Swedish results reproduce the strongest reversal evidence for risky, hard-to-value, and hard-to-arbitrage firms, but mainly after using the macro-orthogonalized sentiment index.

The reported predictive regressions are factor-adjusted throughout. The coefficient on lagged sentiment is a conditional cross-sectional predictability estimate, not a statement that sentiment alone explains most variation in characteristic spreads. This thesis asks whether lagged sentiment remains negatively related to sentiment-prone-minus-insensitive returns after contemporaneous market, size, value, and momentum factor returns are absorbed over the same holding horizon.

5.2.3 Comparison With Baker and Wurgler 2006

The spread evidence inspired by **?** is broadly supportive and concentrated in specific characteristics. Total risk, three-factor IVOL, size, and dividend status behave in the predicted direction and match their main empirical pattern. Low tangibility is one of the strongest Swedish spread results, even though **?** report weaker tangibility evidence. Unprofitable firms and young firms have the expected sign but weaker support. Sales growth is correctly signed but weak. Illiquidity is weak, low profitability has almost no support, and turnover conflicts with the predicted direction.

Table 18: Baker-Wurgler-style spread classification

Characteristic	Mean coef.	Expected-signed t	5% share	Classification
Total risk	-10.21	3.95	1.00	Strong
IVOL (FF3)	-9.50	3.37	1.00	Strong
Low tangibility	-7.33	3.97	1.00	Strong tangibility result
Non-dividend payer	-2.76	2.59	1.00	Strong with dividend-premium caveat
Small firms	-6.09	2.20	0.50	Strong
Unprofitable	-2.25	1.80	0.25	Partial
Young firms	-3.17	1.28	0.25	Partial
Sales growth	-1.58	0.78	0.00	Weak
Illiquidity	-1.06	0.51	0.00	Weak
Low profitability	0.12	0.03	0.00	Weak
Low turnover	1.82	-0.59	0.00	Conflicting

Note. The table uses $SENT^{\perp}$ and averages the four predictive horizons. The 5% share is the share of horizons with a theory-consistent HAC result at the 5% level.

Table 19: Selected 12-month Baker-Wurgler-style coefficients

Characteristic	Coef.	HAC t	Bootstrap t	One-SD effect	Obs.	R^2
Total risk	-19.65	-4.43	-4.18	-10.97	277	0.63
$IVOL^{FF3}$	-18.42	-3.75	-3.76	-10.28	277	0.66
PPE/A	-15.27	-4.83	-4.37	-8.52	277	0.29
Non-dividend payer	-4.96	-2.80	-2.59	-2.77	277	0.68
Unprofitable	-4.86	-3.04	-2.90	-2.71	241	0.68
ME	-14.04	-3.15	-2.84	-7.83	277	0.62
GS	-3.21	-0.61	-0.58	-1.79	229	0.14

Note. The coefficient is on lagged $SENT^{\perp}$. Returns are measured in percentage points. The one-standard-deviation effect uses a sentiment standard deviation of 0.558.

5.3 Stambaugh-Style Leg Decomposition

5.3.1 Aggregate Leg Evidence

? predict that sentiment effects should be strongest in the overpriced side of anomaly strategies. The Swedish leg test is an analogue rather than a replication. Each directional spread is decomposed into the sentiment-prone leg and the insensitive leg. The sentiment-prone leg is the closest analogue to the Stambaugh short leg because it is the side expected to be most exposed to high-sentiment overpricing.

Table 20: Aggregate leg decomposition for the main index

Horizon	Leg	Mean coef.	Mean HAC t	Mean bootstrap t	Median obs.
1	Sentiment-prone leg	-0.49	-0.91	-0.92	288
1	Insensitive leg	0.47	1.39	1.38	288
3	Sentiment-prone leg	-1.13	-0.83	-0.82	286
3	Insensitive leg	0.84	1.18	1.19	286
6	Sentiment-prone leg	-2.84	-1.36	-1.35	283
6	Insensitive leg	1.49	1.28	1.29	283
12	Sentiment-prone leg	-4.95	-1.32	-1.27	277
12	Insensitive leg	2.81	1.59	1.48	277

Note. The sentiment-prone leg is the leg expected to be most exposed to sentiment-driven overpricing.

The aggregate decomposition supports the expected direction but gives a mixed interpretation. The sentiment-prone leg coefficient is negative at every horizon, which is consistent with the Stambaugh-style asymmetry. The insensitive leg coefficient is also positive at every horizon. The spread evidence comes from both weaker subsequent returns for sentiment-prone firms and stronger relative returns for insensitive firms.

5.3.2 Characteristic-Level Leg Evidence

The 12-month leg results separate the clean cases from the weaker cases. Total risk, three-factor IVOL, size, and age have negative and statistically meaningful sentiment-prone-leg coefficients. These characteristics support the Stambaugh-style interpretation most directly. Low tangibility and non-dividend-payers are strong spread results, but their leg decomposition is less clean because the insensitive leg contributes strongly. Sales growth, illiquidity, and low turnover have the expected sentiment-prone-leg sign with weak evidence. Unprofitable firms and low-profitability firms fail the predicted sentiment-prone-leg pattern because their sentiment-prone-leg coefficients are positive.

Table 21: Stambaugh-style 12-month leg classification

Characteristic	Prone coef.	Prone t	Insensitive coef.	Insensitive t	Classification
Total risk	-15.56	-3.50	7.78	3.21	Clean joint support
IVOL (FF3)	-13.43	-2.24	9.07	2.91	Clean joint support
Small firms	-15.28	-3.07	0.62	0.43	Prone leg supports asymmetry
Young firms	-8.85	-2.53	-0.20	-0.07	Prone leg supports asymmetry
Low tangibility	-3.15	-1.01	12.60	5.82	Spread partly driven by insensitive leg
Non-dividend payer	-1.84	-1.22	3.64	2.60	Spread partly driven by insensitive leg
Sales growth	-4.97	-1.57	-1.83	-0.64	Expected sign with weak leg evidence
Illiquidity	-1.25	-0.32	1.99	1.94	Expected sign with weak leg evidence
Low turnover	-4.42	-1.21	-9.64	-3.50	Expected sign with weak leg evidence
Unprofitable	11.32	1.08	3.88	3.33	Fails predicted prone-leg pattern
Low profitability	2.96	1.05	2.97	1.42	Fails predicted prone-leg pattern

Note. Prone denotes the sentiment-prone leg of the Swedish directional spread. Insensitive denotes the comparison leg. Coefficients are from 12-month factor-adjusted regressions on lagged $SENT^{\perp}$.

5.3.3 Comparison With Stambaugh, Yu, and Yuan

Stambaugh, Yu, and Yuan find that sentiment-conditioned anomaly returns are driven mainly by the short leg, where overpricing is hardest to correct. The Swedish evidence only partly follows that pattern. Total risk, three-factor IVOL, and size behave closest to the Stambaugh prediction because the sentiment-prone leg weakens after high sentiment. Age also supports the prone-leg prediction, although it is less central to this thesis results. Low tangibility and dividend status support the BW-style spread prediction more clearly than the Stambaugh-style leg prediction. Sales growth has the expected prone-leg sign but weak evidence. Profitability fails the leg test, and low turnover gives a weak prone-leg sign alongside a negative insensitive-leg coefficient.

5.4 Robustness

HAC/Newey-West standard errors and moving-block bootstrap t-statistics adjust the primary inference for heteroskedasticity, serial correlation, and time-series dependence in overlapping monthly return windows. Those corrections improve the coefficient-level inference, but they do not fully address the pattern that the expected-signed t-statistics rise with the return horizon. Coefficients are naturally larger over longer compounded horizons, but stronger t-statistics at longer horizons raise the possibility that persistent sentiment states and repeated holding-period observations make the $SENT^{\perp}$ evidence look stronger than a strictly non-overlapping design would imply.

Two robustness tests address this concern. First, this thesis estimates non-overlapping horizon regressions. For a return horizon h, the regression sample is split into h calendar-offset subsamples,

$$\mathcal{T}_{h,o} = \{t : m(t) - o \equiv 0 \pmod{h}\}, \qquad o = 0, \ldots, h-1, \tag{17}$$

where $m(t)$ is the monthly time index. The predictive regression is then re-estimated separately in each offset subsample. At the 12-month horizon, this produces twelve regressions for each characteristic, beginning respectively in January, February, and so on. The offset design prevents adjacent observations from mechanically sharing most of the same future holding-period returns. The reported robustness statistics summarize the median coefficient and median expected-signed t-statistic across offsets, together with the fraction of offsets that retain the theory-consistent negative sign.

Table **??** reports the 12-month non-overlapping robustness test, where the concern about repeated holding-period observations is most severe. The main reversal pattern survives this stricter design. Total risk, three-factor idiosyncratic volatility, low tangibility, small firms, unprofitable firms, young firms, and non-dividend payers all retain negative median coefficients and positive expected-signed median t-statistics. The strongest evidence remains concentrated in risk, idiosyncratic volatility, and low tangibility. Sales growth is correctly signed in most offset samples but weak. Low profitability, illiquidity, and low turnover do not provide comparable support. The non-overlapping regressions reduce the concern that the long-horizon results are mechanically driven by repeated overlapping return windows.

Table 22: Persistence robustness: 12-month non-overlapping horizon regressions

Characteristic	Baseline coef.	Baseline t	Non-overlap coef.	Non-overlap t	Sign share
Total risk	-19.65	4.43	-23.41	5.25	1.00
IVOL (FF3)	-18.42	3.75	-23.72	5.59	1.00
Low tangibility	-15.27	4.83	-17.34	4.31	1.00
Small firms	-14.04	3.15	-15.53	3.33	1.00
Unprofitable	-4.86	3.04	-6.99	3.15	1.00
Young firms	-8.37	2.29	-10.02	2.52	1.00
Non-dividend payer	-4.96	2.80	-5.27	2.31	1.00
Sales growth	-3.21	0.61	-3.42	0.44	0.67
Low profitability	0.08	-0.02	-2.06	0.95	0.58
Illiquidity	-2.58	0.72	-3.64	0.80	0.92
Low turnover	4.46	-1.23	4.99	-1.51	0.25

Note. The $SENT^{\perp}$ columns use the overlapping 12-month predictive regressions. The non-overlap columns report the median coefficient and median expected-signed HAC t-statistic across the twelve calendar-offset samples. Sign share is the fraction of offset regressions with the theory-consistent negative coefficient.

Second, this thesis estimates sentiment-innovation regressions. The orthogonalized sentiment index is modelled as a first-order autoregressive process,

$$SENT_t^{\perp} = a + \rho SENT_{t-1}^{\perp} + u_t, \tag{18}$$

and the residual u_t is interpreted as the unexpected component of sentiment after controlling for its own persistence. The predictive regression is then re-estimated with u_{t-1} in place of $SENT^{\perp}_{t-1}$. The specification asks whether new information in sentiment predicts subsequent cross-sectional return spreads, rather than allowing a long high-sentiment regime to enter the regression repeatedly through a highly persistent level variable. These robustness tests do not replace the level specification in **?**, but they provide a direct check on whether the main results are overstated by overlapping returns or persistent sentiment episodes.

The AR(1) coefficient of $SENT^{\perp}$ is 0.794, with an R^2 of 0.634, confirming that sentiment is highly persistent. When the lagged sentiment level is replaced by the lagged AR(1) innovation, the results weaken substantially. The average expected-signed t-statistic is 1.27 at the 12-month horizon, compared with 2.22 in the overlapping level regression. The results are better interpreted as predictability from persistent high- and low-sentiment regimes than from unexpected month-to-month sentiment shocks.

The baseline estimation uses the common-equity universe without an additional market-equity filter. Very small firms are theoretically relevant because sentiment effects are expected to be stronger where valuation is more subjective and arbitrage is more difficult. Market-cap filters were evaluated as robustness checks. The baseline specification instead relies on the common-equity universe filter, return-cleaning rules, point-in-time accounting construction, stale-fundamental caps, lagged portfolio sorts, and robust inference.

The empirical conclusion is supportive but uneven. The spread evidence inspired by **?** supports this thesis hypothesis for risk, IVOL, size, dividend status, and low tangibility. Stambaugh-style leg evidence narrows the strongest conclusion to risk, IVOL, and size. Low tangibility is economically and statistically strong in spreads, but its leg pattern shows that the comparison leg contributes materially. Sales growth, profitability, illiquidity, and turnover should be treated as weak or conflicting evidence rather than supportive findings. The broader question of whether $SENT^{\perp}$ measures sentiment rather than macro-financial conditions is addressed in the discussion section.

6 Discussion

6.1 Sentiment Measurement and Macro-Financial Contamination

? critique the interpretation of the sentiment index in **?** rather than only its forecasting performance. Their evidence accepts the starting point that the index predicts cross-sectional returns, but challenges whether the predictive component is necessarily behavioral sentiment. In their abstract, they state that investor sentiment is “strongly correlated with contemporaneous business cycle variables” and that about 63% of the total variation in the index can be explained by such variables. They further conclude that their results “cast doubt on the notion of ‘sentiment’ as a pure behavioral predictive factor”. Their decomposition separates the fitted business-cycle component, $SENTHAT_t$, from the residual component, $SENTRES_t$. In their return-predictability tests, the business-cycle component carries most of the spread and short-leg predictability, while the residual component is much weaker.

The critique is especially sharp because it applies even to the orthogonalized index in **?**. Baker and

Wurgler residualize their sentiment proxies against a limited macroeconomic control set before extracting the common component. Yet **?** also acknowledge that sentiment measures may be "contaminated by economic fundamentals". The resulting implication is methodological. Orthogonalization against a small set of macro controls improves the interpretation of a composite index, but it does not by itself isolate a pure behavioral factor.

The Swedish common component may combine investor optimism with macro-financial state variation. Survey confidence, turnover, IPO activity, financing composition, and the dividend premium can all reflect sentiment, but they can also move with expected cash flows, discount rates, financing conditions, market liquidity, and risk premia. $SENT^{\perp}$ residualizes the selected proxies against Swedish CPI, PPI, industrial production, and unemployment. That step removes the linear component explained by the selected Swedish macro controls, while leaving open whether other macro-financial information remains in the common component.

6.2 Expanded Macro Orthogonalization

$SENT^{\perp}$ follows the local index-building tradition in **?**, who describe "three basic steps in the construction process": proxy timing, PCA construction, and macroeconomic adjustment. Their stated purpose in the final step is to "control the influence of macroeconomics". This thesis keeps that logic, but adapts the proxy set and return tests to the cross-sectional design in **?**.

The Sibley-expanded Swedish index, $SENT_t^S$, uses the same seven sentiment proxies as $SENT^{\perp}$, including $DIVP_t$, but residualizes them against a broader macro-financial control set before PCA. The expanded controls are unemployment, monthly CPI growth, consumption growth, industrial-production growth, the short rate, the term spread, the value-weighted Swedish market return, monthly market volatility, and the share of zero-return stock-days. The Swedish adaptation uses $UNEMP$, $dCPI$, $dCONS$, $dIND$, $TBILL$, $TERM$, $VWRETD$, $MKTVOL$, and $PCTZERO$.

The Swedish control set is narrower than the full specification in **?** because several U.S. variables do not have clean monthly Swedish equivalents in the available data. Disposable personal income growth, a recession dummy, the default spread, and aggregate dividend yield are omitted because sufficiently consistent Swedish monthly series are not available for those concepts. The test follows the Sibley logic where Swedish data permit a defensible adaptation, while treating the remaining controls as data limitations rather than approximating them with weaker series.

$SENT^{\perp}$ and $SENT^S$ have a correlation of 0.466. The positive correlation shows that they still capture a related market state, while the moderate magnitude shows that the broader macro-financial controls remove a large part of the $SENT^{\perp}$ common component. The comparison tests whether the $SENT^{\perp}$ predictive results survive when the sentiment measure is made more demanding with respect to macro-financial contamination.

6.3 Empirical Consequences

The Sibley-expanded index preserves the expected negative average coefficient at every horizon, but the magnitude and statistical support are much weaker. At the 12-month horizon, the average coefficient falls from -7.89 for $SENT^{\perp}$ to -1.88 for the Sibley-expanded index. The average expected-signed HAC t-statistic falls from 2.22 to 0.73, and the expected-signed bootstrap t-statistic falls from 2.08 to 0.68. The share of characteristics with a theory-consistent 5% HAC result is only 0.09 under the Sibley-expanded index at every horizon.

Table 23: Sibley-style expanded macro orthogonalization by horizon

Horizon	$SENT^{\perp}$ b	$SENT^{\perp}$ HAC	$SENT^{\perp}$ boot	$SENT^{S}$ b	$SENT^{S}$ HAC	$SENT^{S}$ boot	$SENT^{S}$ 5%
1 month	-0.97	1.62	1.62	-0.50	0.75	0.68	0.09
3 months	-2.04	1.46	1.45	-0.49	0.24	0.24	0.09
6 months	-4.37	1.93	1.91	-1.46	0.79	0.75	0.09
12 months	-7.89	2.22	2.08	-1.88	0.73	0.68	0.09

Note. The table averages the eleven directional-spread regressions by horizon. HAC and boot columns report expected-signed t-statistics. Positive values support the sentiment-reversal hypothesis.

The characteristic-level results follow the same pattern. The low-tangibility spread remains the strongest result under the expanded macro adjustment, with a mean expected-signed HAC t-statistic of 2.90. Sales growth and low turnover improve somewhat relative to $SENT^{\perp}$ but remain weak. The main $SENT^{\perp}$ evidence for total risk, $IVOL^{FF3}$, size, non-dividend payers, and unprofitable firms weakens sharply. The mean expected-signed HAC t-statistic falls from 3.37 to 0.43 for $IVOL^{FF3}$, from 2.20 to 0.95 for size, from 2.59 to 0.11 for non-dividend payers, and from 1.80 to 0.09 for unprofitable firms. The broad cross-sectional predictability of $SENT^{\perp}$ is partly sensitive to a more aggressive macro-financial residualization.

The Covid-winsorization check in **????** measures sensitivity to high-leverage crisis observations in the PCA construction. The check leaves the main index unchanged because the Covid shock is treated as a real sentiment episode rather than a data error. Before PCA, the experiment caps only the March 2020 raw ESI_t and CCI_t observations that exceed the four-standard-deviation event-window rule. The baseline $SENT^{\perp}$ series changes little, while the Sibley-expanded series changes materially. The adjustment changes the 12-month Sibley-expanded average coefficient from -1.88 to -6.98, the expected-signed HAC t-statistic from 0.73 to 1.78, and the 5 percent HAC significance share from 0.09 to 0.36. At the 3- and 6-month horizons, the expected-signed HAC t-statistic rises from 0.24 to 1.21 and from 0.79 to 1.72, respectively. The Covid-winsorized evidence shows that the baseline index is not driven by the Covid survey shock, while the Sibley-expanded robustness result is sensitive to crisis-period proxy leverage.

? argue that a fixed PCA index can be restrictive because proxy loadings may vary over time and full-sample PCA weights can embed information from future periods when the index is interpreted as an ex ante predictor. The Covid experiment makes that concern concrete in the Swedish setting. A single crisis-period shock does not materially alter $SENT^{\perp}$, but it has a large effect on the more heavily residualized Sibley PCA. A rolling or recursive PCA design could reduce the influence of one full-sample event and would be closer to a real-time forecasting exercise. This thesis keeps the fixed PCA specification because the empirical purpose is a transparent cross-sectional test based on **?**, not a real-time trading signal. A rolling PCA would introduce additional design choices such as window length, component sign anchoring, minimum observations, and time-varying proxy inclusion, and would be noisy in the Swedish monthly sample. The fixed index is suitable as an in-sample empirical sentiment state, while future research could test whether a rolling PCA version preserves the same cross-sectional return patterns without full-sample PCA weights.

6.4 Implications for Identification

The Sibley-expanded results move this thesis contribution from structural sentiment identification to predictive cross-sectional evidence. $SENT^{\perp}$ predicts Swedish directional spreads in the expected direction, especially for high-risk firms, high-$IVOL^{FF3}$ firms, low-tangibility firms, small firms, and non-dividend payers. The expanded macro adjustment reduces the average 12-month coefficient from -7.89 to -1.88 and lowers the average expected-signed HAC t-statistic from 2.22 to 0.73. A critical reading is that $SENT^{\perp}$ captures a sentiment-related market state with predictive content, while the behavioral component inside that state is only partially isolated.

The main identification problem comes from the economic content of the proxies themselves. Survey confidence, turnover, IPO activity, financing composition, and the dividend premium are plausible sentiment proxies because they move with optimism, speculative participation, hot-issue conditions, issuer timing, and demand for safer dividend-paying firms. The same variables also move with expected cash flows, discount rates, risk appetite, financing conditions, and market liquidity. **?** show that this ambiguity is empirically meaningful in the setting of **?** because a business-cycle component explains much of the variation and predictive content of the sentiment index. The Swedish correlation of 0.466 between $SENT_t^{\perp}$ and $SENT_t^{S}$ points to the same issue. The expanded controls remove a large part of the $SENT^{\perp}$ state variable, and the return evidence weakens with it.

The Sibley-expanded index functions as a stress test of interpretation. A final decomposition of fundamentals and beliefs would require richer identifying variation and additional monthly Swedish data. The Swedish adaptation uses the macro-financial controls that can be measured defensibly, while disposable income growth, a recession dummy, the default spread, and aggregate dividend yield remain outside the available monthly dataset. Some expanded controls may also absorb channels through which sentiment affects prices. Market returns, volatility, liquidity, and term-structure conditions can reflect fundamentals, but they can also be part of the pricing environment in which optimistic demand becomes more or less powerful. The weaker $SENT_t^{S}$ results restrict the interpretation of the $SENT^{\perp}$ evidence and make a purely behavioral reading too strong.

The strongest remaining claim is conditional and cross-sectional. $SENT^{\perp}$ is most informative where the theory predicts larger sentiment effects, namely among firms with subjective valuations or stronger limits to arbitrage. **?** predict larger effects where "valuations are highly subjective and difficult to arbitrage", and the Swedish evidence is strongest for risk, $IVOL^{FF3}$, size, low tangibility, and dividend status. A purely mechanical macro proxy would be expected to map less systematically onto the characteristic families in **?**. The Sibley critique still changes the interpretation. The evidence supports sentiment-related predictability in the Swedish cross-section, while leaving the behavioral purity of the index unresolved.

The leg decomposition narrows the overpricing interpretation further. **?** argue that high sentiment should mainly predict weak returns on the overpriced side because short-sale impediments make overpricing harder to correct. The Swedish leg results fit that logic most clearly for risk, $IVOL^{FF3}$, and size. Low tangibility is strong in the spread regressions, but the comparison leg contributes materially, which makes it weaker evidence for a Stambaugh-style short-leg channel. Profitability, illiquidity, turnover, and sales growth give weaker or conflicting signals. The Swedish evidence supports the spread prediction in **?** more strongly than the stricter Stambaugh overpricing-asymmetry prediction.

The final contribution is a Swedish test of the return-predictability framework in **?** with an explicit Sibley qualification. A locally constructed sentiment-style index predicts future return differences among theoretically sentiment-prone Swedish equities. The predictive state variable combines investor optimism with macro-financial conditions to an extent that remains only partially separated by the current design. This thesis answers whether the Swedish cross-section contains sentiment-related return patterns, while a structural separation of behavioral beliefs from macro-financial state variation remains outside the identification power of the analysis.

7 Conclusion

8 Perspective

A Appendix

A.1 Raw Sentiment-Proxy Fundamentals

Table 24: Descriptive statistics for raw sentiment proxy inputs

Variable	Mean	Std. dev.	Min.	Max.	Median	Obs.
Economic sentiment index (ESI_t)	0.0007	0.0397	-0.1412	0.4921	-0.0011	299
Consumer confidence index (CCI_t)	0.0008	0.0273	-0.0653	0.1290	-0.0019	299
Turnover ($TURN_t$)	-3.1224	0.3964	-4.0663	-1.9195	-3.1493	299
IPO number ($NIPO_t$)	1.4181	2.6715	0.0000	21.0000	0.0000	299
IPO initial return ($RIPO_t$)	0.1503	0.5412	-0.6511	4.6848	0.0622	130
Equity-to-debt ratio (ED_RATIO_t)	0.2929	0.0791	0.2114	0.5483	0.2558	299
Dividend premium ($DIVP_t$)	-0.3791	0.7499	-2.1240	0.8512	-0.2626	289

Note. Statistics are computed from the unstandardized monthly proxy series before PCA and before macro residualization. $RIPO_t$ is observed only in months with observed IPO initial-return information. $NIPO_t$ equals zero in months without IPO activity. $TURN_t$ and $DIVP_t$ are logarithmic proxy constructions.

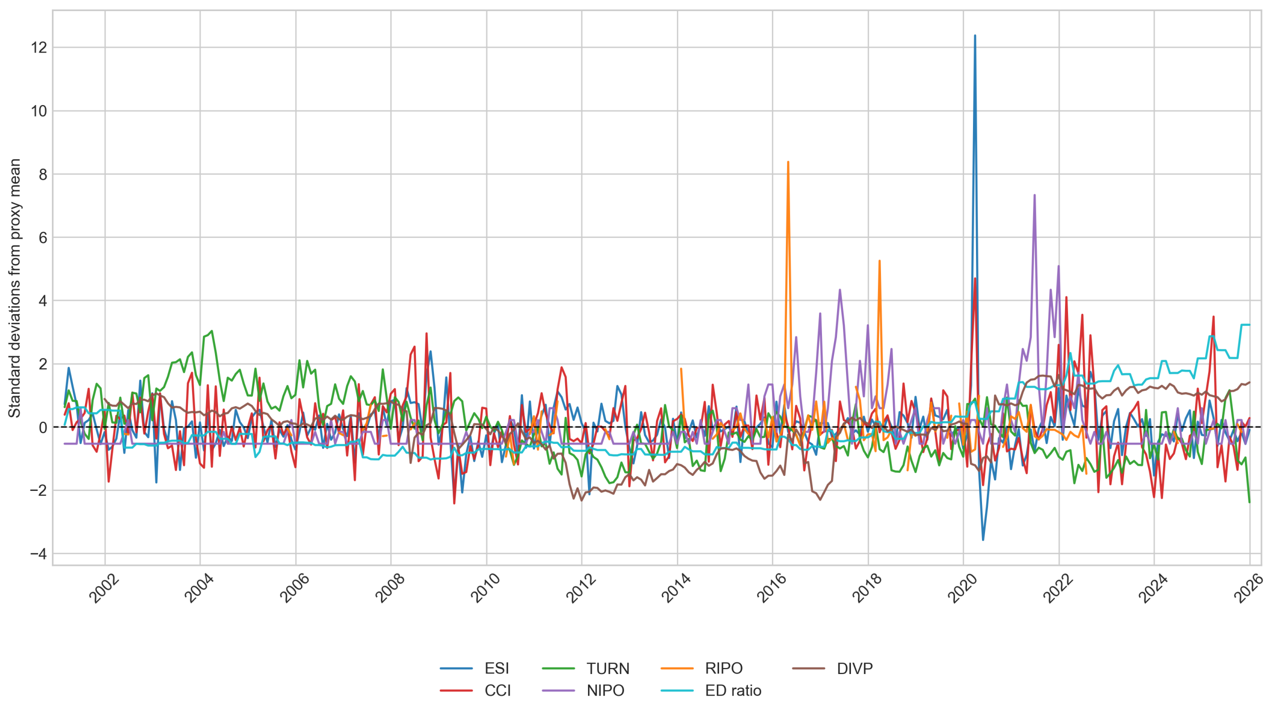

Figure 3: Standardized sentiment proxy inputs over time

Note. Each proxy is standardized by subtracting its sample mean and dividing by its sample standard deviation. The horizontal dashed line marks zero. $RIPO_t$ is plotted only in months with observed IPO initial-return information.

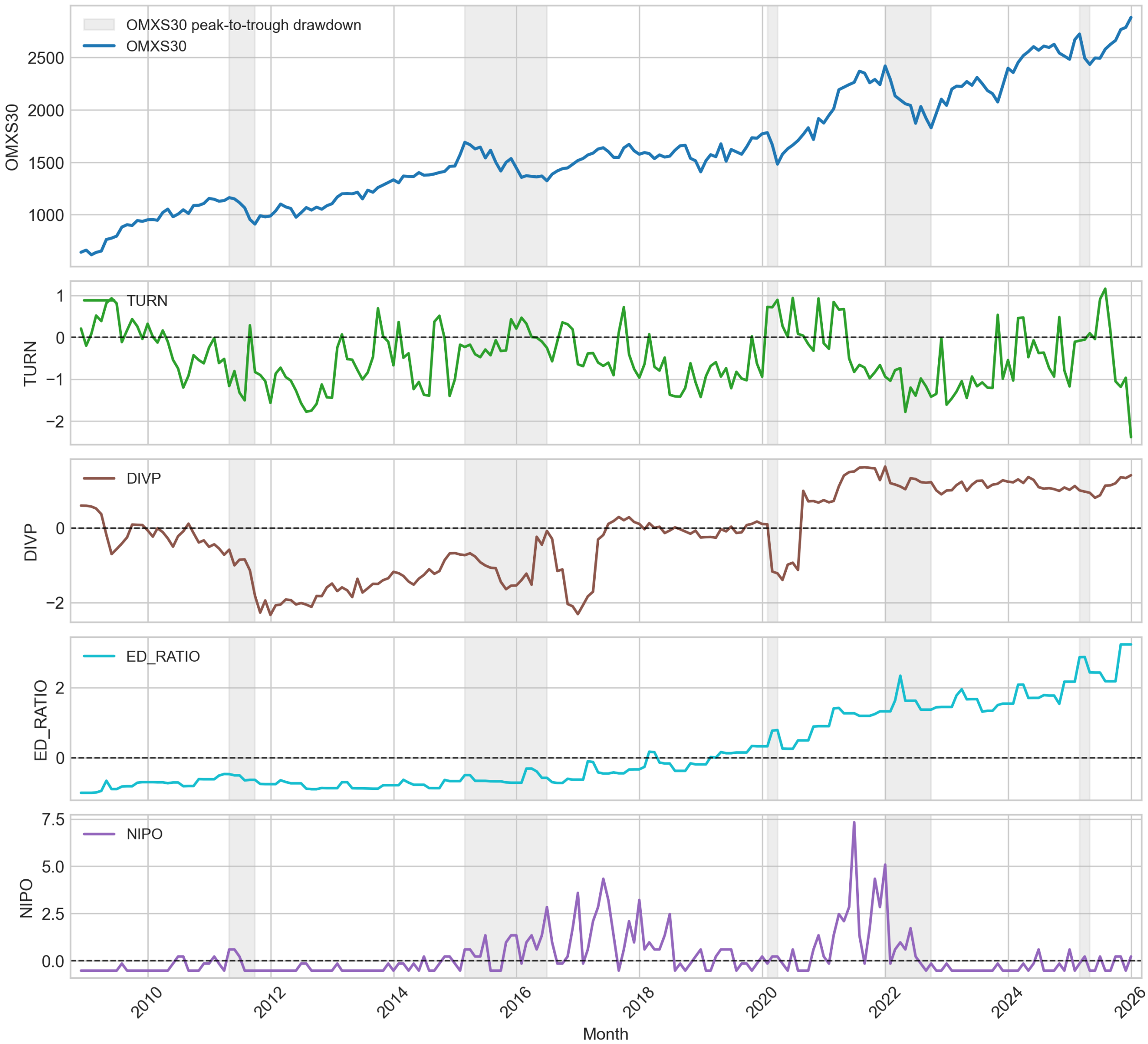

Figure 4: OMXS30 drawdowns and selected standardized sentiment proxies

Note. Grey areas mark the five largest peak-to-trough OMXS30 drawdowns in the monthly OMXS30 series. Proxy series are standardized by subtracting each proxy mean and dividing by its sample standard deviation. The horizontal dashed line in each proxy panel marks zero.

A.2 Sibley-Expanded Sentiment Robustness by Characteristic and Horizon

The comparison below reports the coefficient and HAC t-statistic for each directional-spread regression under $SENT^{\perp}$ and $SENT^{S}$. The status column classifies each characteristic-horizon test by whether the expected negative coefficient is significant at the 5 percent HAC level before and after the Sibley-expanded macro adjustment. Negative coefficients support the expected sentiment-reversal pattern because each spread is coded as sentiment-prone minus sentiment-insensitive.

Table 25: Sibley-expanded sentiment robustness by characteristic and horizon

Characteristic	$SENT^{\perp}$		$SENT^{S}$		Status
	Coef.	HAC t	Coef.	HAC t	
1-month horizon					
Small firms	-1.34	-1.90	-0.79	-1.09	Not 5% before
Young firms	-0.38	-0.80	-0.07	-0.17	Not 5% before
Total risk	**-2.60**	**-3.50**	**-1.14**	**-1.64**	**Loses 5%**
$IVOL^{FF3}$	**-2.24**	**-2.69**	**-0.62**	**-0.81**	**Loses 5%**
Low profitability	-0.44	-0.64	0.24	0.44	Not 5% before
Unprofitable	-0.54	-0.96	-0.15	-0.34	Not 5% before
Non-dividend payer	**-0.63**	**-2.24**	**0.07**	**0.38**	**Loses 5%**
Low tangibility	-1.27	-2.99	-1.55	-2.50	Stays 5%
Sales growth	-0.80	-1.33	-0.53	-0.97	Not 5% before
Illiquidity	-0.30	-0.67	0.15	0.31	Not 5% before
Low turnover	-0.11	-0.16	-1.08	-1.86	Not 5% before
3-month horizon					
Small firms	-2.44	-1.38	0.01	0.01	Not 5% before
Young firms	-1.06	-0.79	-0.46	-0.48	Not 5% before
Total risk	**-6.49**	**-3.56**	**-1.30**	**-0.96**	**Loses 5%**
$IVOL^{FF3}$	**-5.60**	**-2.97**	**-0.63**	**-0.41**	**Loses 5%**
Low profitability	0.43	0.34	1.23	1.43	Not 5% before
Unprofitable	-1.11	-1.36	0.48	1.22	Not 5% before
Non-dividend payer	**-1.88**	**-2.46**	**0.30**	**0.74**	**Loses 5%**
Low tangibility	-4.46	-3.86	-3.11	-2.88	Stays 5%
Sales growth	-0.88	-0.64	-0.90	-0.98	Not 5% before
Illiquidity	-0.03	-0.03	0.76	1.12	Not 5% before
Low turnover	1.04	0.62	-1.82	-1.45	Not 5% before
6-month horizon					
Small firms	**-6.55**	**-2.39**	**-3.03**	**-1.27**	**Loses 5%**
Young firms	-2.86	-1.23	-0.54	-0.39	Not 5% before
Total risk	**-12.08**	**-4.30**	**-3.29**	**-1.62**	**Loses 5%**
$IVOL^{FF3}$	**-11.74**	**-4.05**	**-1.44**	**-0.72**	**Loses 5%**
Low profitability	0.41	0.18	1.62	0.97	Not 5% before
Unprofitable	-2.50	-1.84	0.17	0.26	Not 5% before
Non-dividend payer	**-3.58**	**-2.88**	**-0.20**	**-0.28**	**Loses 5%**
Low tangibility	-8.34	-4.21	-4.96	-3.10	Stays 5%
Sales growth	-1.41	-0.55	-1.62	-1.11	Not 5% before
Illiquidity	-1.32	-0.65	-0.16	-0.15	Not 5% before
Low turnover	1.91	0.67	-2.67	-1.25	Not 5% before
12-month horizon					
Small firms	**-14.04**	**-3.15**	**-6.64**	**-1.46**	**Loses 5%**
Young firms	**-8.37**	**-2.29**	**-1.18**	**-0.52**	**Loses 5%**
Total risk	**-19.65**	**-4.43**	**-1.63**	**-0.46**	**Loses 5%**
$IVOL^{FF3}$	**-18.42**	**-3.75**	**0.59**	**0.22**	**Loses 5%**
Low profitability	0.08	0.02	1.27	0.70	Not 5% before
Unprofitable	**-4.86**	**-3.04**	**-1.32**	**-1.50**	**Loses 5%**
Non-dividend payer	**-4.96**	**-2.80**	**-1.39**	**-1.27**	**Loses 5%**
Low tangibility	-15.27	-4.83	-8.97	-3.11	Stays 5%
Sales growth	-3.21	-0.61	-2.49	-1.24	Not 5% before
Illiquidity	-2.58	-0.72	1.09	0.62	Not 5% before
Low turnover	4.46	1.23	-0.03	-0.01	Not 5% before

Note. Coefficients are percentage-point effects on future directional-spread returns. HAC t-statistics use the Newey-West standard errors from the predictive regressions. Status compares theory-consistent 5 percent HAC significance under $SENT^{\perp}$ and $SENT^{S}$. Bold rows identify cases where significance is lost under the Sibley-expanded index.

A.3 Covid-Winsorized Sibley Robustness

The Covid-winsorized appendix check modifies only the Sibley discussion-layer robustness index. The main index remains the unadjusted $SENT^{\perp}$ series because the Covid shock is treated as an economically real sentiment episode rather than as a data error. The figure compares the original and Covid-winsorized sentiment series. $SENT^{\perp}$ changes little, while the more aggressively residualized Sibley-expanded index is materially affected by the crisis-period proxy adjustment.

Figure 5: Original and Covid-winsorized sentiment indices

Note. The Covid-winsorized series cap extreme March–April 2020 raw proxy observations before PCA construction. The adjustment is triggered by the March 2020 ESI_t and CCI_t observations. The baseline $SENT^{\perp}$ series remains close to the original series, while $SENT^S$ changes materially.

The full characteristic-by-horizon predictive regressions for the Covid-winsorized Sibley-expanded index are reported below. Relative to the unadjusted Sibley-expanded index, the Covid-winsorized version stays significant in 4 characteristic-horizon tests, gains significance in 11, and is never significant in 29. Relative to $SENT^{\perp}$, 12 tests stay significant, 8 lose significance, 3 gain significance, and 21 are never significant. The adjustment makes the Sibley-expanded index more predictive in several rows, especially at longer horizons, but it does not fully reproduce the $SENT^{\perp}$ cross-sectional pattern.

Table 26: Covid-winsorized Sibley-expanded sentiment regressions by characteristic and horizon

Characteristic	Hor.	Coef.	HAC t	Boot. t	Exp. HAC	Exp. boot	Unadj. Sibley → adj. Sibley	$SENT^{\perp}$ → adj. Sibley
Small firms	1m	-1.29	-1.78	-1.73	1.78	1.73	Never sig.	Never sig.
Young firms	1m	-0.25	-0.48	-0.47	0.48	0.47	Never sig.	Never sig.
Total risk	1m	-1.36	-1.73	-1.67	1.73	1.67	Never sig.	Lost sig.
$IVOL^{FF3}$	1m	-1.50	-1.71	-1.68	1.71	1.68	Never sig.	Lost sig.
Low profitability	1m	0.43	1.05	0.98	-1.05	-0.98	Never sig.	Never sig.
Unprofitable	1m	0.36	1.00	0.99	-1.00	-0.99	Never sig.	Never sig.
Non-dividend payer	1m	-0.10	-0.30	-0.30	0.30	0.30	Never sig.	Lost sig.
Low tangibility	1m	-1.29	-2.95	-2.77	2.95	2.77	Stayed sig.	Stayed sig.
Sales growth	1m	-1.10	-1.79	-1.65	1.79	1.65	Never sig.	Never sig.
Illiquidity	1m	-0.09	-0.19	-0.18	0.19	0.18	Never sig.	Never sig.
Low turnover	1m	-0.60	-0.86	-0.87	0.86	0.87	Never sig.	Never sig.
Small firms	3m	-4.07	-2.36	-2.28	2.36	2.28	Gained sig.	Gained sig.
Young firms	3m	-0.36	-0.24	-0.23	0.24	0.23	Never sig.	Never sig.
Total risk	3m	-4.56	-2.76	-2.60	2.76	2.60	Gained sig.	Stayed sig.
$IVOL^{FF3}$	3m	-4.56	-2.39	-2.29	2.39	2.29	Gained sig.	Stayed sig.
Low profitability	3m	1.28	1.19	1.12	-1.19	-1.12	Never sig.	Never sig.
Unprofitable	3m	0.32	0.34	0.34	-0.34	-0.34	Never sig.	Never sig.
Non-dividend payer	3m	-0.77	-0.90	-0.93	0.90	0.93	Never sig.	Lost sig.
Low tangibility	3m	-4.45	-4.13	-3.93	4.13	3.93	Stayed sig.	Stayed sig.
Sales growth	3m	-2.72	-2.03	-1.95	2.03	1.95	Gained sig.	Gained sig.
Illiquidity	3m	-0.22	-0.18	-0.17	0.18	0.17	Never sig.	Never sig.
Low turnover	3m	0.22	0.11	0.11	-0.11	-0.11	Never sig.	Never sig.
Small firms	6m	-10.07	-3.78	-3.62	3.78	3.62	Gained sig.	Stayed sig.
Young firms	6m	-2.37	-0.93	-0.91	0.93	0.91	Never sig.	Never sig.
Total risk	6m	-7.40	-2.59	-2.45	2.59	2.45	Gained sig.	Stayed sig.
$IVOL^{FF3}$	6m	-8.27	-2.60	-2.52	2.60	2.52	Gained sig.	Stayed sig.
Low profitability	6m	2.12	1.00	0.95	-1.00	-0.95	Never sig.	Never sig.
Unprofitable	6m	-0.59	-0.37	-0.37	0.37	0.37	Never sig.	Never sig.
Non-dividend payer	6m	-1.70	-1.19	-1.19	1.19	1.19	Never sig.	Lost sig.
Low tangibility	6m	-9.31	-5.58	-5.38	5.58	5.38	Stayed sig.	Stayed sig.
Sales growth	6m	-4.42	-2.04	-2.01	2.04	2.01	Gained sig.	Gained sig.
Illiquidity	6m	-2.08	-0.88	-0.87	0.88	0.87	Never sig.	Never sig.
Low turnover	6m	0.13	0.04	0.04	-0.04	-0.04	Never sig.	Never sig.
Small firms	12m	-20.10	-4.71	-4.29	4.71	4.29	Gained sig.	Stayed sig.
Young firms	12m	-6.71	-1.55	-1.45	1.55	1.45	Never sig.	Lost sig.
Total risk	12m	-15.30	-3.31	-3.26	3.31	3.26	Gained sig.	Stayed sig.
$IVOL^{FF3}$	12m	-13.69	-2.53	-2.48	2.53	2.48	Gained sig.	Stayed sig.
Low profitability	12m	3.74	1.22	1.21	-1.22	-1.21	Never sig.	Never sig.
Unprofitable	12m	-1.27	-0.63	-0.60	0.63	0.60	Never sig.	Lost sig.
Non-dividend payer	12m	-2.77	-1.16	-1.11	1.16	1.11	Never sig.	Lost sig.
Low tangibility	12m	-16.49	-5.71	-5.04	5.71	5.04	Stayed sig.	Stayed sig.
Sales growth	12m	-7.34	-1.90	-1.72	1.90	1.72	Never sig.	Never sig.
Illiquidity	12m	-2.25	-0.53	-0.55	0.53	0.55	Never sig.	Never sig.
Low turnover	12m	5.43	1.26	1.22	-1.26	-1.22	Never sig.	Never sig.

Note: The table reports directional-spread predictive regressions using the Covid-winsorized Sibley-expanded sentiment index. Expected-signed t-statistics multiply the reported sentiment coefficient t-statistic by the sign predicted by the sentiment-reversal hypothesis, so positive values support the hypothesis. The two shift columns classify whether a theory-consistent 5% HAC result stayed significant, lost significance, gained significance, or was never significant when moving from the comparison index to the Covid-winsorized Sibley index. The Covid treatment caps only extreme raw proxy observations in March–April 2020 before PCA construction and is used only as discussion-layer robustness evidence.

A.4 Preliminary Sentiment PCA Eigenvectors

Table 27: Full preliminary sentiment PCA eigenvectors for lead-lag treatment

Variable	Comp1	Comp2	Comp3	Comp4	Comp5	Comp6	Comp7
ESI_t	0.0239	-0.1690	0.4822	0.0454	-0.4779	-0.2561	0.0666
ESI_{t-1}	0.0057	-0.1595	0.4295	-0.2846	0.4711	0.4206	0.0647
CCI_t	0.0453	-0.1840	0.4554	0.1684	-0.4807	0.3450	-0.0295
CCI_{t-1}	0.0357	-0.2021	0.4733	-0.1591	0.4419	-0.4450	0.0413
$TURN_t$	-0.1036	0.5228	0.2277	0.2387	0.0920	0.0346	0.1341
$TURN_{t-1}$	-0.0899	0.5062	0.2332	0.2809	0.1040	0.1266	0.2189
$NIPO_t$	0.1538	-0.2916	-0.0574	0.6044	0.1586	0.0419	0.3599
$NIPO_{t-1}$	0.1621	-0.3165	-0.0415	0.5402	0.2506	-0.0080	-0.3544
$RIPO_t$	-0.0243	0.0012	-0.0141	0.0125	0.0853	0.5552	-0.0514
$RIPO_{t-1}$	-0.0200	0.0085	-0.0650	0.0583	0.0273	-0.2294	0.6000
ED_RATIO_t	0.5091	-0.0319	-0.0875	-0.1714	-0.0699	0.1231	0.2846
ED_RATIO_{t-1}	0.5103	-0.0348	-0.0816	-0.1776	-0.0383	0.1293	0.2800
$DIVP_t$	0.4533	0.2746	0.0927	0.0618	0.0360	-0.0953	-0.2776
$DIVP_{t-1}$	0.4487	0.2808	0.1139	0.0433	0.0223	-0.1399	-0.2621

Note. The table reports eigenvectors for the seven retained components from the preliminary PCA on current and one-month-lagged values of the seven sentiment proxies. The retained components explain 86.0% of total standardized proxy variance.

A.5 Raw Compustat Input-Field Coverage

Table 28: Raw Compustat input-field coverage

Title	Compustat Code	Raw-row availability	Coverage (%)
Monthly market fields			
Price Close - Monthly	prccm	256,671	96.99%
Trading Volume - Monthly	cshtrm	211,442	79.90%
Daily market, return, and liquidity fields			
Price Close - Daily	prccd	6,307,856	99.99%
Shares Outstanding - Company	cshoc	6,189,018	98.10%
Trading Volume - Daily	cshtrd	4,279,555	67.84%
Total Return Factor	trfd	6,308,333	99.99%
Scale and revenue fields			
Assets - Total	atq	61,856	94.29%
Sales/Turnover (Net)	saleq	49,286	75.13%
Sales/Turnover (Net), YTD	saley	53,105	80.95%
Revenue - Total	revtq	58,650	89.40%
Tangibility and investment fields			
Property Plant and Equipment - Total (Net)	ppentq	52,759	80.42%
Capital Expenditures	capxy	36,080	55.00%
Earnings fields			
Income Before Extraordinary Items	ibq	61,071	93.09%
Net Item - Total	nitq	66	0.1%
Income Before Extraordinary Items, YTD	iby	65,418	99.72%
Net Item - Total, YTD	nity	452	0.69%
Book-equity and balance-sheet fields			
Common/Ordinary Equity - Total	ceqq	51,639	78.72%
Stockholders Equity > Parent > Index Fundamental > Quarterly	seqq	63,087	96.17%
Stockholders Equity - Total	teqq	49,564	75.55%
Liabilities - Total	ltq	61,836	94.26%
Preferred/Preference Stock (Capital) - Total	pstkq	498	0.76%
Retained Earnings	req	34,678	52.86%
Dividend fields			
Cash Dividends	dvy	12,475	19.02%
Dividends - Total	dvty	6,213	9.47%
Dividends - Total, quarterly	dvtq	4,861	7.41%
External-finance fields			
Sale of Common and Preferred Stock	sstky	18,083	27.56%
Purchase of Common and Preferred Stock	prstkcy	926	1.41%
Long-Term Debt - Issuance	dltisy	5	0.01%
Fields not present in the extract			
Research and Development Expense	not present	0	0%
Deferred Taxes and Investment Tax Credit	not present	0	0%

Note. Coverage is measured before point-in-time monthly carry-forward and before portfolio formation. Accounting-field coverage uses the raw quarterly fundamentals denominator of 65,602 firm-quarter records identified by unique gvkey/datadate pairs. Monthly market-field coverage uses 264,645 raw security-month rows from the monthly price extract. Daily market-field coverage uses 6,308,719 raw security-day rows from the daily price extract. Variables marked as not present were not included in the saved WRDS extract used for this thesis.